# Supplementary Material
# Beyond Instance-Wise Early Classification: Adaptive EEG Acquisition with Stopping-Dependent History for Sleep Staging

This document provides the supplementary analyses and reproducibility details for the accompanying PAUSE manuscript. Sections, figures, tables, equations, and references are numbered independently of the main manuscript. All notation follows the main manuscript unless explicitly defined below.

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GUIDE TO THE SUPPLEMENTARY MATERIAL.


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## I. Additional Related-Work Discussion

### A. Resource-Efficient Sleep Staging

Resource-efficient sleep staging has primarily been approached from the model side. TinySleepNet, LWSleepNet, Micro SleepNet, and MorpheusNet reduce parameter count, computation, or inference latency for deployment on resource-constrained devices [1]–[4]. These methods improve processing efficiency but generally retain the conventional assumption that a complete, fixed-duration EEG epoch has already been acquired. Consequently, reducing model complexity does not necessarily reduce the amount of EEG collected by the sensing front end.

MASS shifts the efficiency objective from computation to signal acquisition by learning from partially and irregularly observed EEG [5]. Related mini-epoch studies also investigate classification from shortened, fixed-duration inputs [6]. These approaches demonstrate that reliable staging does not always require all 30 seconds of an epoch. However, their observation lengths or masking ratios are specified independently of the collected EEG signal. Any variation in the resulting allocation is therefore exogenous rather than a response to epoch difficulty. In MASS the retained patches are moreover scattered rather than contiguous, and its epoch-level masking may drop an epoch in its entirety, so acquisition is committed in advance and some labels are produced without any signal from the epoch they describe.

Reducing the number of acquired samples is also distinct from making a label available earlier. Many context-aware sleep-staging models operate on predefined windows or complete epoch representations, so a prediction is produced only after the prescribed input has become available. For example, MASS constructs a global prompt from a complete 32-epoch window and uses a bidirectional inter-epoch encoder, so no label in a window can be emitted before its last epoch has been acquired. It therefore reduces partial-observation cost without by itself providing evidence-dependent acquisition or variable-latency decisions, and its outputs remain retrospective rather than available for during-sleep intervention.

### B. Early Classification and Finite-Horizon Stopping

Early classification of time series (ECTS) predicts from progressively observed prefixes while trading classification performance against observation delay [7], [8]. Existing methods differ mainly in how they trigger a decision. EARLIEST jointly learns a recurrent classifier and a stochastic halting policy [9]. TEASER accepts a prediction after a reliability classifier and temporal-consistency test deem it sufficiently trustworthy, while the Second-order Confidence Network learns stopping signals from both the current prediction and its preceding probability trajectory [10], [11]. ECONOMY-$\gamma$-Max estimates the future cost of waiting within confidence groups, and EarlyMix improves classification from information-limited prefixes through hierarchical mixing [12], [13].

Sequential density-ratio estimation provides a complementary, decision-theoretic route. LSEL learns multiclass sequential evidence for MSPRT-based early classification and is designed to remain discriminative under class imbalance [14]; it supplies an evidence-learning objective rather than, by itself, a finite-horizon stopping rule. FIRMBOUND subsequently formulates early classification within a fixed horizon as backward induction over immediate and future Bayes risks [15]. Its statistically consistent construction estimates both sufficient statistics and conditional expectations, while its faster GP alternative trades that consistency route for lower fitting cost. PAUSE follows the latter finite-horizon regression pattern.

Accordingly, its GP stopper is a fitted approximation to the continuation-risk function, rather than a direct inheritance of FIRMBOUND's full consistency or optimality guarantee.

### C. Cross-Epoch Context under Adaptive Acquisition

Automatic sleep staging differs from standard ECTS because its classification instances are physiologically interdependent. SleepContextNet and L-SeqSleepNet show that temporal context across consecutive epochs captures sleep-stage continuity and transition structure that may be unavailable from an isolated epoch [16], [17]. These models, however, assume that the representations of surrounding epochs are constructed from complete or externally specified observations.

Conventional ECTS can model temporal dependence within an individual time series, but it generally assigns a stopping time to each instance without allowing that decision to determine the observations available to later instances. In adaptive sleep staging, stopping epoch $m$ at $\tau_m$ makes its unacquired suffix unavailable when classifying subsequent epochs. A model trained only with complete historical epochs therefore encounters a policy-induced history shift during deployment. The relevant gap is not the absence of temporal modeling in ECTS or sleep staging, but the absence of a mechanism linking an epoch's stopping decision to the historical evidence visible thereafter. PAUSE addresses this gap by coupling adaptive prefix acquisition with policy-compatible cross-epoch modeling, using only the current acquired prefix and endpoint summaries formed from previously acquired signals.

## II. Dataset Description

### A. Dataset Amount

We evaluate PAUSE on four public sleep-staging benchmarks. DREAMS-SUB comprises 20 overnight recordings from 20 healthy adults aged 20–65 years [18]. Sleep-EDF-20 and Sleep-EDF-78 are two commonly used subsets of the Sleep-EDF Expanded database: the former contains 39 recordings from 20 subjects, whereas the latter contains 153 recordings from 78 subjects [19]. The Sleep Heart Health Study (SHHS) is a large multi-center cohort originally designed to study the cardiovascular consequences of sleep-disordered breathing [20], [21]. Following the cohort selection adopted in prior single-channel sleep-staging work [22], we use a subset of 329 recordings, each from a different subject.

Although the original polysomnography recordings contain multiple physiological channels, all experiments use one EEG derivation per dataset. Specifically, we use Cz-A1 for DREAMS-SUB, Fpz-Cz for both Sleep-EDF subsets, and C4-A1 for SHHS. The original sampling rates are 200 Hz for DREAMS-SUB, 100 Hz for Sleep-EDF, and 125 Hz for SHHS. DREAMS-SUB and SHHS are therefore downsampled to 100 Hz, giving a unified 3,000-sample waveform for every 30-s epoch.

We harmonize the annotations into the five AASM stages W, N1, N2, N3, and REM [23]. UNKNOWN epochs are removed, and the original N3 and N4 labels are merged into N3. To prevent long wake periods from dominating the class distribution, leading and trailing wakefulness are each restricted to the 30 minutes immediately adjacent to the scored sleep period. After this processing, the four datasets contain 20,242, 40,293, 162,456, and 324,148 epochs, respectively; their class distributions are reported in Table II.

### B. Data Preprocessing

Our preprocessing converts each 30-s EEG epoch into a sequence of 30 spectral patches, one for each second. It consists of three steps: splitting the waveform into one-second segments, transforming each segment into the frequency domain, and applying logarithmic scaling. All selected EEG channels are represented at 100 Hz, yielding 3,000 samples per epoch. Sleep-EDF is natively sampled at 100 Hz, DREAMS-SUB is downsampled from 200 Hz during dataset preparation, and SHHS is resampled from 125 Hz with a rational polyphase factor of $4/5$.

Let $\mathbf{x}_n \in \mathbb{R}^{3000}$ denote the 100-Hz waveform of epoch $n$. We divide it into 30 non-overlapping one-second segments. For patch $p \in \{0, \dots, 29\}$ and within-patch sample $m \in \{0, \dots, 99\}$, the windowed signal is

$$s_{n,p}[m] = x_n[100p + m]\, w[m],$$
$$w[m] = 0.54 - 0.46 \cos\left(\frac{2\pi m}{99}\right).$$

where $w[m]$ is the non-periodic 100-sample Hamming window used in our implementation. Each segment is zero-padded to 256 samples before applying the discrete Fourier transform:

$$X_{n,p}[q] = \sum_{m=0}^{99} s_{n,p}[m] \exp\left(-\mathrm{i}\frac{2\pi q m}{256}\right).$$

Because the input signal is real-valued, the negative-frequency half of the transform is redundant. We therefore consider only the 129 non-negative-frequency coefficients indexed by $q = 0, \dots, 128$. The first coefficient, $q = 0$, is the DC component and represents the segment's mean level rather than oscillatory EEG activity. We remove this coefficient and retain $q = 1, \dots, 128$. At a sampling rate of 100 Hz, their frequencies are $f_q = 100q/256$ Hz, spanning approximately 0.39–50 Hz. The stored feature is the log magnitude

$$S_{n,p,q} = 20 \log_{10}\left(|X_{n,p}[q]| + 10^{-8}\right), \qquad \mathbf{S}_n \in \mathbb{R}^{30\times 128}.$$

We refer to $\mathbf{S}_n$ as `PSD100`. Since $20 \log_{10} |X| = 10 \log_{10} |X|^2$, it is the log periodogram of the segment up to a patch- and frequency-independent additive constant, which the standardization below absorbs. This construction preserves a one-to-one correspondence between patch $p$ and the associated one-second acquisition interval, so an acquired prefix of length $k$ is represented by the first $k$ rows of $\mathbf{S}_n$.

For each outer fold, we compute one scalar mean $\mu_{\mathrm{tr}}$ and standard deviation $\sigma_{\mathrm{tr}}$ over all entries of the training-fold tensors used for optimization, and apply the same statistics to both training and held-out folds:

$$\widetilde{S}_{n,p,q} = \frac{S_{n,p,q} - \mu_{\mathrm{tr}}}{\sigma_{\mathrm{tr}}}.$$

TABLE II
DISTRIBUTION OF SLEEP STAGES ACROSS FOUR DATASETS.

| Dataset | W | N1 | N2 | N3 | REM | Total |
|---|---|---|---|---|---|---|
| **DREAMS-SUB** | 3559 (17.58%) | 1480 (7.31%) | 8251 (40.76%) | 3933 (19.43%) | 3019 (14.91%) | 20242 |
| **Sleep-EDF-20** | 6558 (16.28%) | 2766 (6.86%) | 17629 (43.75%) | 5629 (13.97%) | 7711 (19.14%) | 40293 |
| **Sleep-EDF-78** | 35788 (22.03%) | 20342 (12.52%) | 67661 (41.65%) | 12894 (7.94%) | 25771 (15.86%) | 162456 |
| **SHHS** | 45613 (14.07%) | 10304 (3.18%) | 142125 (43.85%) | 54690 (16.87%) | 71416 (22.03%) | 324148 |

The standardized files also retain the raw $1 \times 3{,}000$ waveform for baselines that operate directly in the time domain.

### C. Cross-Validation and Out-of-Fold Protocol

We use subject-wise 20-fold cross-validation for all four datasets. DREAMS-SUB contains exactly 20 subjects and is evaluated by leave-one-subject-out cross-validation, with one subject held out in each fold. For Sleep-EDF-20 and Sleep-EDF-78, all recordings from the same subject are first grouped using the subject identifier in the file name, after which the subjects are shuffled with seed 2025 and partitioned into 20 folds. Consequently, each Sleep-EDF-20 fold contains one subject, while the Sleep-EDF-78 folds contain either three or four subjects. SHHS contains one recording per selected subject; its 329 recordings are shuffled with the same seed and divided into nine folds of 17 subjects and 11 folds of 16 subjects. No subject appears in both the training and held-out portions of a fold.

These subject assignments are fixed once and reused without modification in all reported experiments, including PAUSE, its ablations, the system-level ECTS baselines, the frozen-backbone decision-policy comparison, and the full-observation sleep-staging baselines. Consequently, every method is evaluated on the same held-out subjects in each fold to ensure fairness.

For each outer fold, the PAUSE backbone is trained on the other 19 folds. The scalar normalization statistics and empirical class prior are computed only from that fold's training data and are then reused for its held-out data. Cross-epoch windows are constructed separately within each data file before different files are concatenated, so neither a window nor its endpoint history crosses a file boundary.

Calibration and stopper fitting use cross-fitted out-of-fold (OOF) predictions. We index folds by a superscript so that the subscripts of the main manuscript keep denoting an epoch and a prefix length. Let $f^{(i)}$ be the backbone trained without outer fold $i$, and let $\mathbf{s}_{n,k}^{(i)}$ denote its evidence for a held-out epoch $n$ of fold $i$ at prefix length $k$. For a target fold $v$, we exclude all predictions and labels from fold $v$ and pool the OOF records $\{(\mathbf{s}_{n,k}^{(i)}, y_n) : i \neq v\}$ from the remaining 19 folds. A separate positive temperature $T_k^{(v)}$ is fitted for every prefix length by minimizing negative log-likelihood, first averaging over the cached predictions within each source fold and then weighting the 19 source folds equally. Each source fold uses the class prior estimated from its own training partition during fitting; at evaluation time, $T_k^{(v)}$ is applied to the target-fold evidence while the target model's training-fold prior $\widehat{\boldsymbol{\pi}}^{(v)}$ remains outside the temperature scaling:

$$\boldsymbol{\pi}_{n,k}^{\mathrm{cal}} = \mathrm{softmax}\left(\frac{\mathbf{s}_{n,k}^{(v)}}{T_k^{(v)}} + \log \widehat{\boldsymbol{\pi}}^{(v)}\right).$$

For the paired-history model, each OOF cache is generated from 32 reproducible random-history draws (seeds 2025–2056), and duplicated epochs introduced when completing the final 32-epoch window are removed before fitting. The same leave-target-fold-out OOF caches are used to fit the fold-specific GP stoppers, so neither the temperatures nor the stoppers of fold $v$ use fold $v$ as a fitting source.

More precisely, the OOF construction asserts exclusion at the sample and label level: the fitting sample of $T_k^{(v)}$ and of the GP stopper of fold $v$ contains no epoch and no label of fold $v$, and the backbone $f^{(v)}$ evaluated on fold $v$ never observed fold $v$ during training. Because each source backbone $f^{(i)}$ excludes only its own fold $i$, the procedure is cross-fitted rather than fully nested: every temperature and every stopper is estimated from logits that are held out with respect to the folds on which they are computed. Cross-fitting is the appropriate construction for fitting a second-stage estimator on first-stage predictions. The in-sample alternative—fitting $T^{(v)}$ from the predictions of $f^{(v)}$ on its own training folds—would replace held-out logits with in-sample ones, which are systematically overconfident and would bias the fitted temperature downward. The subsection *Sensitivity of the fitted temperature to its source set* quantifies how far $T_k$ moves when an entire source fold is deleted from the fitting sample.

## III. EVALUATION METRICS

The main manuscript reports accuracy (ACC), macro-F1 (MF1), the macro geometric mean (MGm), class-wise F1, the mean acquisition length $\mathbb{E}[\tau]$, and the average matched-budget Pareto gain. This section gives the computation of each quantity, together with the two aggregation conventions used across folds, so that every reported number can be reconstructed from the retained per-condition summaries.

### A. Notation and Confusion Counts

This section keeps the notation of the main manuscript: $C$ sleep stages, $P$ patches per epoch, $k$ a prefix length, and $\tau_n \in \{1, \ldots, P\}$ the acquisition endpoint of epoch $n$. Two conventions are local to the evaluation setting. First, $n$ indexes the $N_{\mathrm{ev}}$ pooled held-out epochs of a dataset rather than the epochs of a single window. Second, $\widehat{y}_n$ abbreviates the stage that the evaluated system finally outputs for epoch $n$, which

for PAUSE is $\widehat{y}_{n,\tau_n}$. With $y_n \in \{1, \ldots, C\}$ the reference stage, all stage-level metrics are functions of the confusion counts

$$N_{c,a} = \sum_{n=1}^{N_{\mathrm{ev}}} \mathbb{I}[y_n = c]\ \mathbb{I}[\widehat{y}_n = a]\,,$$

whose row index $c$ is the reference stage and whose column index $a$ is the predicted stage, following the convention of the cost matrix $\Lambda_{c,a}$ of the main manuscript. We write

$$N_c = \sum_{a=1}^{C} N_{c,a}, \qquad \widehat{N}_c = \sum_{r=1}^{C} N_{r,c},$$

for the held-out support of stage $c$ and the number of epochs predicted as stage $c$, so that $N_{\mathrm{ev}} = \sum_c N_c$. The support $N_c$ is the evaluation-side counterpart of the training-side class count $M_c$; the latter is reserved for the empirical prior $\hat{\boldsymbol{\pi}}$ and the cost weights $w_c$.

### B. Classification Metrics

Accuracy is the pooled fraction of correctly staged epochs,

$$\mathrm{ACC} = \frac{1}{N_{\mathrm{ev}}} \sum_{c=1}^{C} N_{c,c}.$$

The per-stage precision, recall, and F1 score are

$$\mathrm{Pre}_c = \frac{N_{c,c}}{\widehat{N}_c}, \qquad \mathrm{Rec}_c = \frac{N_{c,c}}{N_c},$$

$$\mathrm{F1}_c = \frac{2\,\mathrm{Pre}_c \mathrm{Rec}_c}{\mathrm{Pre}_c + \mathrm{Rec}_c} = \frac{2N_{c,c}}{N_c + \widehat{N}_c},$$

where the second form is the one evaluated in code, since it stays well defined when a stage is never predicted. The class-wise F1 columns, and the N1 curves in the lower row of the speed–accuracy figure of the main manuscript, are $\mathrm{F1}_c$ for a single stage. Macro-F1 averages them with equal weight per stage,

$$\mathrm{MF1} = \frac{1}{C} \sum_{c=1}^{C} \mathrm{F1}_c,$$

so a stage contributes independently of its prevalence. This is the property that makes MF1 the primary axis of the frontier: N1 covers only 3.2%–12.5% of the epochs, and a prevalence-weighted score would leave a stopper free to abandon it.

The macro geometric mean pairs recall with specificity. With $\mathrm{TN}_c = N_{\mathrm{ev}} - N_c - \widehat{N}_c + N_{c,c}$ and $\mathrm{FP}_c = \widehat{N}_c - N_{c,c}$,

$$\mathrm{Spe}_c = \frac{\mathrm{TN}_c}{\mathrm{TN}_c + \mathrm{FP}_c} = \frac{N_{\mathrm{ev}} - N_c - \widehat{N}_c + N_{c,c}}{N_{\mathrm{ev}} - N_c},$$

$$\mathrm{MGm} = \frac{1}{C} \sum_{c=1}^{C} \sqrt{\mathrm{Spe}_c\, \mathrm{Rec}_c}.$$

MGm is therefore the arithmetic mean over stages of a per-stage geometric mean of sensitivity and specificity, not the geometric mean of the recalls. The two macro scores respond differently to false positives: $\mathrm{Spe}_c$ carries the denominator $N_{\mathrm{ev}} - N_c$, so spurious predictions of a rare stage move it far less than they move $\mathrm{Pre}_c$ and hence $\mathrm{F1}_c$. MF1 and MGm can consequently order two models differently, as they do for PAUSE and AttnSleep on Sleep-EDF-20, and we therefore report both.

### C. Prefix-Wise Curves

For a specified history condition and prediction rule, the prefix-wise curve $\mathrm{MF1}(k)$ applies the macro-F1 definition above to the confusion counts obtained from predictions at prefix length $k$. In the controlled fixed-prefix reference, every evaluation epoch is assigned the same endpoint $k$, and the endpoints visible to later epochs are fixed to $k$ as well.

The class-balanced paired-history LSEL objective is designed to learn prior-free class evidence. More precisely, the pairwise differences of $s_{n,k}$ play the role of class-conditional log-density ratios, while the empirical class prior $\widehat{\pi}$ is introduced only at the decision layer. For backbone and calibration diagnostics, we distinguish the following three rules:

$$\widehat{y}_{n,k}^{\mathrm{ML}} = \arg\max_c\ s_{n,k}(c), \tag{1}$$

$$\widehat{y}_{n,k}^{\mathrm{MAP}} = \arg\max_c\ \pi_{n,k}^{\mathrm{cal}}(c) \tag{2}$$

$$= \arg\max_c \left\{ \frac{s_{n,k}(c)}{T_k} + \log \widehat{\pi}_c \right\}, \tag{3}$$

$$\widehat{y}_{n,k}^{\mathrm{MAP}_1} = \arg\max_c \left\{ s_{n,k}(c) + \log \widehat{\pi}_c \right\}. \tag{4}$$

Here $\widehat{y}^{\mathrm{ML}}$ is the prior-free evidence rule used for checkpoint selection; $\widehat{y}^{\mathrm{MAP}}$ is the calibrated, prior-aware rule used by the fixed-prefix and specified score-based comparisons; and $\widehat{y}^{\mathrm{MAP}_1}$ is its uncalibrated $T_k = 1$ reference. Each reported curve uses one rule throughout, although curves from different rules may be contrasted explicitly. Because $T_k > 0$ rescales all class evidence scores equally, it leaves the ML prediction unchanged but can change the prior-aware MAP prediction.

The utility-aware PAUSE stopper is distinct from these diagnostic rules: its final prediction minimizes class-weighted posterior risk and therefore need not coincide with MAP when the misclassification costs are class dependent.

### D. Acquisition Metrics

Acquisition is measured in acquired patches, one patch per second, so $\mathbb{E}[\tau]$ is simultaneously the mean number of decisions and the mean acquisition duration in seconds of a 30-second epoch. The reported quantities are

$$\mathbb{E}[\tau] = \frac{1}{N_{\mathrm{ev}}} \sum_{n=1}^{N_{\mathrm{ev}}} \tau_n = \frac{1}{N_{\mathrm{ev}}} \sum_{c=1}^{C} N_c\, \mathbb{E}[\tau \mid c],$$

$$\mathbb{E}[\tau \mid c] = \frac{1}{N_c} \sum_{n:\, y_n = c} \tau_n,$$

so the overall mean is the support-weighted average of the stage-conditional means. Conditioning is on the reference stage $y_n$, not on the prediction, which keeps $\mathbb{E}[\tau \mid c]$ interpretable when a rule systematically mislabels a stage. The overall horizontal axis of the speed–accuracy figures and the subscripts of the ablation table are $\mathbb{E}[\tau]$; the class-wise panels

use $\mathbb{E}[\tau \mid c]$ against $\mathrm{F1}_c$, so a curve is read at the budget that stage actually consumes. Derived statements use

$$\text{observed fraction} = \mathbb{E}[\tau]/P,$$
$$\text{discarded fraction} = 1 - \mathbb{E}[\tau]/P,$$

and the acquisition reduction between two operating points is their ratio $\mathbb{E}[\tau]_{\mathrm{ref}}/\mathbb{E}[\tau]_{\mathrm{PAUSE}}$, reported as a factor.

### *E. Speed–Accuracy Frontier and Matched-Budget Pareto Gain*

Each evaluated system exposes a single scalar that trades acquisition against accuracy: the cost $\eta$ for PAUSE and FIRMBOUND, the penalty $\lambda$ for EARLIEST, the confidence-group parameter for ECONOMY-$\gamma$-Max, and the decision threshold for EarlyMix. Sweeping that scalar for one system yields a set of operating points, each pairing a realized acquisition budget $B$ with the score attained there. The budget is the mean acquisition length, $B = \mathbb{E}[\tau]$, or $B = \mathbb{E}[\tau \mid c]$ in a class-wise panel. A point is dominated when another point of the same system costs no more and scores no less, with at least one inequality strict. The Pareto front collects the non-dominated points in increasing $B$.

For method $m$, let $[B_m^{\min}, B_m^{\max}]$ denote the native budget support of its observed Pareto front, and let $\Phi_m(B)$ be the piecewise-linear interpolation of that front within this interval. When constructing the comparator envelope, we use the budget-feasible extension

$$\widetilde{\Phi}_m(B) = \begin{cases} \text{undefined}, & B < B_m^{\min}, \\ \Phi_m(B), & B_m^{\min} \le B \le B_m^{\max}, \\ \Phi_m(B_m^{\max}), & B > B_m^{\max}. \end{cases}$$

The right extension reflects that an operating point costing $B_m^{\max}$ remains feasible under any larger budget. It is used only to construct the comparator envelope and does not extend the integration domain beyond the native PAUSE front.

Write $\Phi_{\mathrm{P}}$ for the PAUSE front and let $\mathcal{C}$ denote the set of comparators. At budget $B$, the set of comparators with at least one feasible operating point is

$$\mathcal{U}(B) = \left\{ m \in \mathcal{C} : B \ge B_m^{\min} \right\}.$$

Their upper envelope is

$$\Phi^{\star}_{\mathrm{base}}(B) = \max_{m \in \mathcal{U}(B)} \widetilde{\Phi}_m(B),$$

and the matched-budget Pareto gap is

$$\Delta_{\mathrm{Pareto}}(B) = \Phi_{\mathrm{P}}(B) - \Phi^{\star}_{\mathrm{base}}(B).$$

Reading $\Phi^{\star}_{\mathrm{base}}(B)$ as the best interpolated score available from any comparator without exceeding budget $B$ makes $\Delta_{\mathrm{Pareto}}(B)$ a like-for-like comparison at equal acquisition rather than a comparison between arbitrarily selected operating points. The shaded band of the speed–accuracy figures is the region between $\Phi_{\mathrm{P}}$ and $\Phi^{\star}_{\mathrm{base}}$, drawn in the accent colour where $\Delta_{\mathrm{Pareto}}(B) \ge 0$ and in grey where $\Delta_{\mathrm{Pareto}}(B) < 0$.

We also define the comparable integration interval as

$$B_{\min} = \max\left\{ B_{\mathrm{P}}^{\min}, \min_{m \in \mathcal{C}} B_m^{\min} \right\}, \qquad B_{\max} = B_{\mathrm{P}}^{\max}.$$

Thus, PAUSE is evaluated only within the native support of its observed Pareto front, while at least one comparator has a feasible operating point throughout $[B_{\min}, B_{\max}]$. In particular, the interval is not extended to the physical limit $P$ unless $B_{\mathrm{P}}^{\max} = P$. For the pooled macro-F1 panels, the resulting intervals, rounded to three decimals, are $[1.147, 17.011]$ for DREAMS, $[1.102, 15.916]$ for Sleep-EDF-20, $[1.125, 17.679]$ for Sleep-EDF-78, and $[1.224, 21.098]$ for SHHS. The class-wise panels apply the same rule using $B = \mathbb{E}[\tau \mid c]$ and therefore generally have different integration intervals.

The annotated average Pareto gain is the signed mean gap over this interval:

$$\overline{\Delta}_{\mathrm{Pareto}} = \frac{100}{B_{\max} - B_{\min}} \int_{B_{\min}}^{B_{\max}} \Delta_{\mathrm{Pareto}}(B)\, \mathrm{d}B,$$

where the factor 100 converts a difference between $[0,1]$-valued scores into percentage points. The integral is evaluated by the trapezoidal rule on a grid $B_{\min} = B_1 < \cdots < B_J = B_{\max}$, writing $\Delta_i = \Delta_{\mathrm{Pareto}}(B_i)$:

$$\overline{\Delta}_{\mathrm{Pareto}} = \frac{100}{B_{\max} - B_{\min}} \sum_{i=1}^{J-1} \frac{\Delta_i + \Delta_{i+1}}{2} \left(B_{i+1} - B_i\right).$$

The PAUSE front and comparator envelope are piecewise linear. A grid containing all of their breakpoints therefore evaluates the integral exactly, and the fine uniform grid used in our calculation agrees with this value beyond the reported precision. Because the reported quantity is a signed mean, intervals on which a comparator is ahead reduce the average gain. If the positive region is considered separately, the same expression can instead be applied to

$$[\Delta_{\mathrm{Pareto}}(B)]_+ = \max\{\Delta_{\mathrm{Pareto}}(B), 0\}.$$

The annotated value, such as the $+13.73$ macro-F1 percentage points reported in the figure, refers to the signed mean over the entire comparable interval rather than to the positive region alone or to a single favourable operating point.

### *F. Aggregation across Folds*

The speed–accuracy frontiers and acquisition ablations pool confusion counts $N_{c,a}$ and stopping times $\tau_i$ over the 20 held-out folds before computing each metric. Every held-out epoch contributes equally, so folds containing more epochs have greater weight; pooling also stabilizes minority-stage metrics at aggressive budgets. The full-observation sleep-staging comparisons instead compute each metric within each fold and report the equal-weight mean over the 20 folds. The main-paper full-observation table additionally reports the population standard deviation, $\sigma = \sqrt{\frac{1}{20}\sum_{f=1}^{20}(m_f - \overline{m})^2}$ (ddof$=0$), matching the archived baseline summaries. These standard deviations describe between-fold variation, not standard errors or paired significance tests. For nonlinear metrics such as macro-F1 and MGm, the fold mean generally differs from the metric computed from pooled counts.

### G. Deployment Cost Metrics

Computational cost is reported per stopping decision, that is per acquired patch under incremental decoding, and counted as multiply–accumulate operations (MACs) from the analytic operation count of the exported single-step graph; the corresponding FLOP count is twice the MAC count. Parameter memory assumes single-precision storage. Latency is measured per decision on a single CPU thread in thermal steady state, and reported as the median together with the 90th and 99th percentiles of the measured runs. The compute duty cycle of an epoch follows from the median per-decision latency $t_{\mathrm{dec}}$ under the 1 Hz acquisition regime, since exactly one decision is evaluated per acquired patch and acquisition halts at $\tau$:

$$\text{duty cycle} = \frac{\mathbb{E}[\tau] \cdot t_{\mathrm{dec}}}{30\ \mathrm{s}}.$$

## IV. PAUSE Structure and Hyperparameters

This section expands the PAUSE architecture of the main manuscript and keeps its notation: a window holds $L = 32$ consecutive epochs of $P = 30$ one-second patches, $n$ indexes the epoch being classified and $m < n$ a preceding epoch of the same window, $k$ is the current prefix length, $\tau_m$ the acquisition endpoint of epoch $m$, $d$ the token width, $H$ the number of heads, $d_h = d/H$ the head dimension, $\beta$ the rotary base, and $C$ the number of sleep stages. We additionally write $\widetilde{\mathbf{S}}_{n,k} \in \mathbb{R}^Q$, $Q = 128$, for the standardized `PSD100` spectrum of patch $k$ of epoch $n$, that is, the $k$-th row of the tensor $\widetilde{\mathbf{S}}_n$ of the preceding section. Throughout, $\sigma(z) = z \cdot \mathrm{sigmoid}(z)$ is the SiLU nonlinearity, $\odot$ the elementwise product, and bias terms are omitted for readability.

Tables III–VII report the complete configuration of PAUSE. For all methods used in this paper, we control the random seed as 2025.

### A. Causal Endpoint-Memory Backbone

The backbone maps each acquired spectral patch through a gated embedding and a left-causal stem, then refines the resulting token with seven causal Transformer blocks. At prefix $k$, attention is restricted to the current-epoch tokens $1{:}k$ and one retained endpoint from each preceding epoch in the window. A shared head converts the final state into the prior-free class evidence $\mathbf{s}_{n,k}$ available at every prefix. The following sections detail these stages and the paired-history objective used to train them. Tables III and IV describe the hyperparameters and training settings of the Causal Endpoint-Memory Backbone.

*1) Patch embedding.:* Every patch is lifted to width $d = 128$ by a gated (SwiGLU) projection,

$$\mathbf{g}_{n,k} = \mathbf{E}_d\, \mathrm{LN}\Big(\sigma\Big(\mathbf{E}_g \widetilde{\mathbf{S}}_{n,k}\Big) \odot \mathbf{E}_u \widetilde{\mathbf{S}}_{n,k}\Big)\,,$$

with $\mathbf{E}_g, \mathbf{E}_u \in \mathbb{R}^{d\times Q}$, $\mathbf{E}_d \in \mathbb{R}^{d\times d}$, and LN a LayerNorm. The branch $\mathbf{E}_u \widetilde{\mathbf{S}}_{n,k}$ carries the projected spectrum and the gate $\sigma(\mathbf{E}_g \widetilde{\mathbf{S}}_{n,k})$ rescales it channelwise, so the patch is encoded by an input-dependent rather than a fixed linear map.

TABLE III
Architecture of the causal endpoint-memory backbone.

| Hyperparameter | Value |
|---|---|
| input patch dimension $Q$ | 128 |
| patches per epoch $P$ | 30 |
| cross-epoch window length $L$ | 32 |
| token dimension $d$ | 128 |
| patch embedding | gated (SwiGLU) MLP |
| causal patch stem | enabled |
| patch-stem left context | 4 patches |
| Transformer depth | 7 |
| attention heads $H$ | 8 |
| head dimension $d_h$ | 16 |
| attention span | causal prefix + endpoint memory |
| positional encoding | 2D RoPE |
| RoPE base $\beta$ | 30 |
| FFN hidden width | $4d = 512$ |
| activation function | SwiGLU |
| normalization | pre-norm RMSNorm |
| dropout rate | 0.2 |
| number of sleep stages $C$ | 5 |
| trainable parameters | 2.0 M |

*2) Causal patch stem.:* The left-causal convolutional stem then mixes each token with its four predecessors and yields the token $\mathbf{z}_{n,k} \in \mathbb{R}^d$ of the main manuscript. With per-channel depthwise weights $\boldsymbol{\phi}_0, \ldots, \boldsymbol{\phi}_4 \in \mathbb{R}^d$ and the convention $\mathbf{g}_{n,k} \equiv \mathbf{0}$ for $k \le 0$,

$$\mathbf{u}_{n,k} = \sum_{\delta=0}^{4} \boldsymbol{\phi}_\delta \odot \mathrm{LN}(\mathbf{g}_{n,k-\delta})\,,$$
$$\mathbf{z}_{n,k} = \mathbf{g}_{n,k} + \mathbf{D}_d(\sigma(\mathbf{D}_g \mathbf{u}_{n,k}) \odot \mathbf{D}_u \mathbf{u}_{n,k})\,,$$

where $\mathbf{D}_g, \mathbf{D}_u \in \mathbb{R}^{2d\times d}$ and $\mathbf{D}_d \in \mathbb{R}^{d\times 2d}$. The kernel spans five patches but is padded on the left only, so $\mathbf{z}_{n,k}$ depends on patches $k-4, \ldots, k$ and never on an unacquired one.

*3) Endpoint-memory attention.:* The window is processed by seven pre-norm blocks with $H = 8$ heads of dimension $d_h = 16$. Writing $\mathbf{h}_{n,k}^{(t)}$ for the state after block $t$, we set $\mathbf{h}_{n,k}^{(0)} = \mathbf{z}_{n,k}$ and identify the output of the last block with the hidden state of the main manuscript, $\mathbf{h}_{n,k} = \mathbf{h}_{n,k}^{(7)}$. Each block projects its input to the per-head queries, keys and values and rotates $\mathbf{q}$ and $\mathbf{k}$ with the two-dimensional RoPE. Specifically, the patch and epoch subspaces of a head each hold $d_h/2 = 8$ coordinates, and $\mathcal{R}_p(k;\beta)$ and $\mathcal{R}_e(n;\beta)$ rotate their $d_h/4$ coordinate pairs by the angles $k\,\beta^{-4i/d_h}$ and $n\,\beta^{-4i/d_h}$ respectively, for $i = 0, \ldots, d_h/4 - 1$, with base $\beta = 30$. Position therefore enters only through relative offsets, and no learned positional table is indexed by prefix length, so one set of weights serves every $k$.

The endpoint memory is realized inside every block rather than by a separate module. Block $t$ reuses the rotated key and the value that a preceding epoch $m$ has already computed at

its own endpoint,

$$\widetilde{\mathbf{k}}^{a,(t)}_{m,\tau_m}, \qquad \mathbf{v}^{a,(t)}_{m,\tau_m},$$

which are exactly the endpoint columns of the candidate banks $\overline{\mathbf{K}}^a_n$ and $\overline{\mathbf{V}}^a_n$ of the main manuscript. Cross-epoch context is thus carried at all seven depths at a cost of one key–value pair per preceding epoch, and the mask $\mathcal{A}\big((n,k),(m,j)\big)$ remains the only route by which one epoch reaches another. The state $\mathbf{h}^{(t)}_{n,k}$ is a function of $\widetilde{\mathbf{S}}_{n,1:k}$ and of the endpoint memories of earlier epochs alone, so no decision consumes evidence that has not been acquired; and stopping epoch $m$ fixes $\tau_m$, which is precisely the column that later epochs read.

*4) Blocks and classification head.:* Let MHA denote the attention above, with its per-head outputs $\mathbf{o}^a_{n,k}$ concatenated and linearly projected, and let $\mathrm{FFN}(\mathbf{x}) = \mathbf{F}_d(\sigma(\mathbf{F}_g\mathbf{x}) \odot \mathbf{F}_u\mathbf{x})$ be the SwiGLU feed-forward network of hidden width $4d = 512$. Block $t$ applies

$$\widetilde{\mathbf{h}}^{(t)}_{n,k} = \mathbf{h}^{(t)}_{n,k} + \mathrm{MHA}\Big(\mathrm{RMSNorm}\Big(\mathbf{h}^{(t)}_{n,k}\Big)\Big),$$
$$\mathbf{h}^{(t+1)}_{n,k} = \widetilde{\mathbf{h}}^{(t)}_{n,k} + \mathrm{FFN}\Big(\mathrm{RMSNorm}\Big(\widetilde{\mathbf{h}}^{(t)}_{n,k}\Big)\Big),$$

for $t = 0, \ldots, 6$, with dropout $0.2$ on the attention weights, on the output projections and inside the gated branches. The shared evidence head of the main manuscript is applied to the normalized final state,

$$\mathbf{s}_{n,k} = \mathbf{W}_s\,\mathrm{RMSNorm}(\mathbf{h}_{n,k}) + \mathbf{b}_s \in \mathbb{R}^C,$$

where RMSNorm uses $\varepsilon = 10^{-6}$. The backbone has 2.0 M trainable parameters.

*5) Optimization.:* Training follows Table IV. The objective is the paired-history LSEL $\mathcal{L}_{\mathrm{paired}}$ of the main manuscript, whose outer normalization $\frac{1}{C}\sum_c \frac{1}{M_c}$ keeps the minority stages from being absorbed by the dominant class. Both history views $\boldsymbol{\tau}^{\mathrm{F}}$ and $\boldsymbol{\tau}^{\mathrm{R}}$ of every window are concatenated along the minibatch dimension and optimized in one update, so the number of sequences per step is twice the window batch size listed in the table. Following the time-averaged criterion of FIRMBOUND, the retained checkpoint maximizes the prefix-averaged macro-F1 $\frac{1}{P}\sum_{k=1}^{P}\mathrm{MF1}(k)$.

TABLE IV
TRAINING CONFIGURATION OF THE CAUSAL ENDPOINT-MEMORY BACKBONE. THE WINDOW BATCH SIZE IS 16 FOR DREAMS-SUB AND SLEEP-EDF-20 AND 32 FOR SLEEP-EDF-78 AND SHHS; PAIRED-HISTORY TRAINING DOUBLES THE NUMBER OF SEQUENCES PER OPTIMIZATION STEP.

| Setting | Value |
|---|---|
| objective | $\mathcal{L}_{\mathrm{paired}}$ (LSEL) |
| class balancing | $\frac{1}{C}\sum_c \frac{1}{M_c}$ |
| history views | $\boldsymbol{\tau}^{\mathrm{F}}, \boldsymbol{\tau}^{\mathrm{R}}$ |
| optimizer | Lion |
| peak learning rate | $3 \times 10^{-4}$ |
| minimum learning rate | $1 \times 10^{-6}$ |
| weight decay | 0.05 |
| learning-rate schedule | cosine with warmup |
| warmup epochs | 10 |
| maximum training epochs | 100 |
| early-stopping patience | 40 epochs |
| gradient-norm clipping | 1.0 |
| windows per batch | 16 or 32 |
| numerical precision | FP32 |
| checkpoint criterion | $\frac{1}{P}\sum_k \mathrm{MF1}(k)$ |
| random seed | 2025 |

## B. Hyperparameter Search

We selected the backbone configuration through a low-fidelity, multi-objective Optuna search on DREAMS-SUB. Each trial trained the paired-history model for 50 epochs while retaining the 100-epoch cosine-schedule horizon, and was searched using the first five LOSO folds from the full 20-subject data pool. We optimized two complementary objectives. The fold-averaged $\mathrm{MF1}(30)$ measures the accuracy ceiling attained when all 30 patches are acquired, whereas the fold-averaged prefix-averaged macro-F1, $\frac{1}{P}\sum_{k=1}^{P}\mathrm{MF1}(k)$, measures predictive quality throughout the acquisition trajectory. This prevents gains at $k = 30$ from compromising the intermediate prefixes used for training and stopping.

Table V lists the search space. Every dimension is categorical, including the learning rate and the weight decay, which are drawn from explicit grids rather than from continuous ranges; the sampler therefore proposes only values that appear in the table. Its single conditional dependency is the rotary base, whose candidate set differs between the two positional variants because the 1D encoding indexes a flattened sequence of $L \times P = 960$ positions, whereas the 2D encoding indexes at most $P = 30$ patches and $L = 32$ epochs and so requires a much shorter wavelength range. Counting that dependency once rather than as a product, the space contains 9,307,872 configurations. The quantities that define the acquisition protocol rather than the model are held fixed across all trials: the window length $L = 32$, the horizon $P = 30$, the input width $Q = 128$, the causal patch stem with its four-patch left context, the window batch size 16, the Lion optimizer, the cosine schedule with 10 warmup epochs and minimum learning rate $10^{-6}$, and the training seed 2025.

We jointly maximized the two objectives without reducing them to a weighted scalar. Consequently, the search does not produce a uniquely ordered best trial, but a Pareto front of configurations for which neither objective can be improved without degrading the other. The search used a multivariate grouped TPE sampler with 20 startup trials and no pruner, and completed 261 trials; Fig. 1 shows the resulting objective values. Among the distinct configurations on the Pareto front, we selected the model with the smallest parameter count as the default PAUSE configuration. This choice contains 1,998,981 trainable parameters (approximately 2.0 M). It therefore preserves a Pareto-optimal balance between full-epoch accuracy and performance across intermediate prefixes while favoring the compact model required for deployment.

Fig. 2 reports the importance of each search dimension, computed separately for the two objectives over the 261 completed trials. The plot covers the twelve unconditional

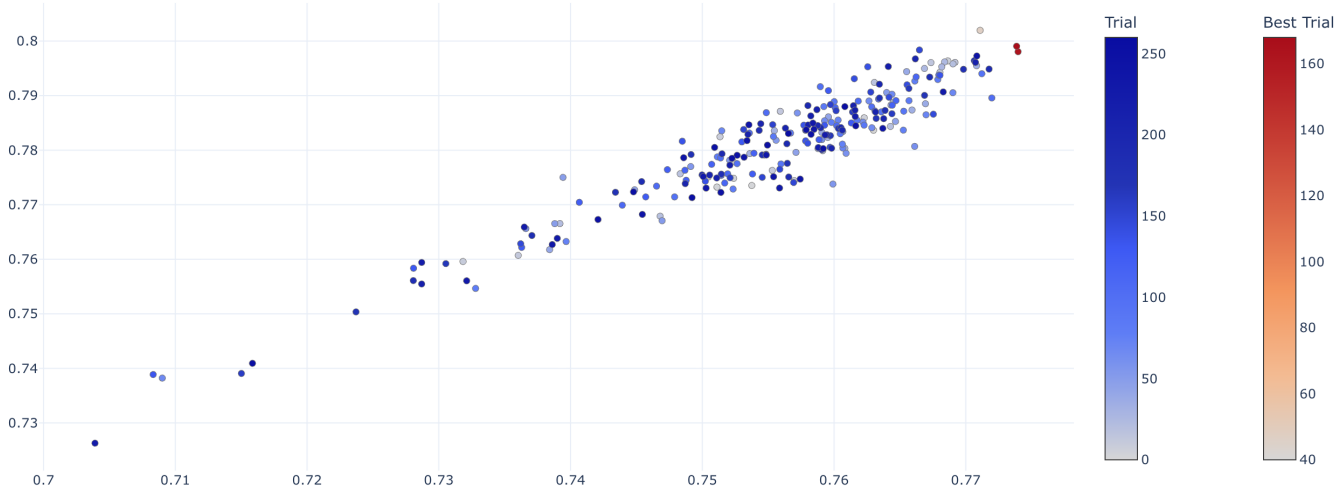



Fig. 1. Optuna selection on hyperparameters.

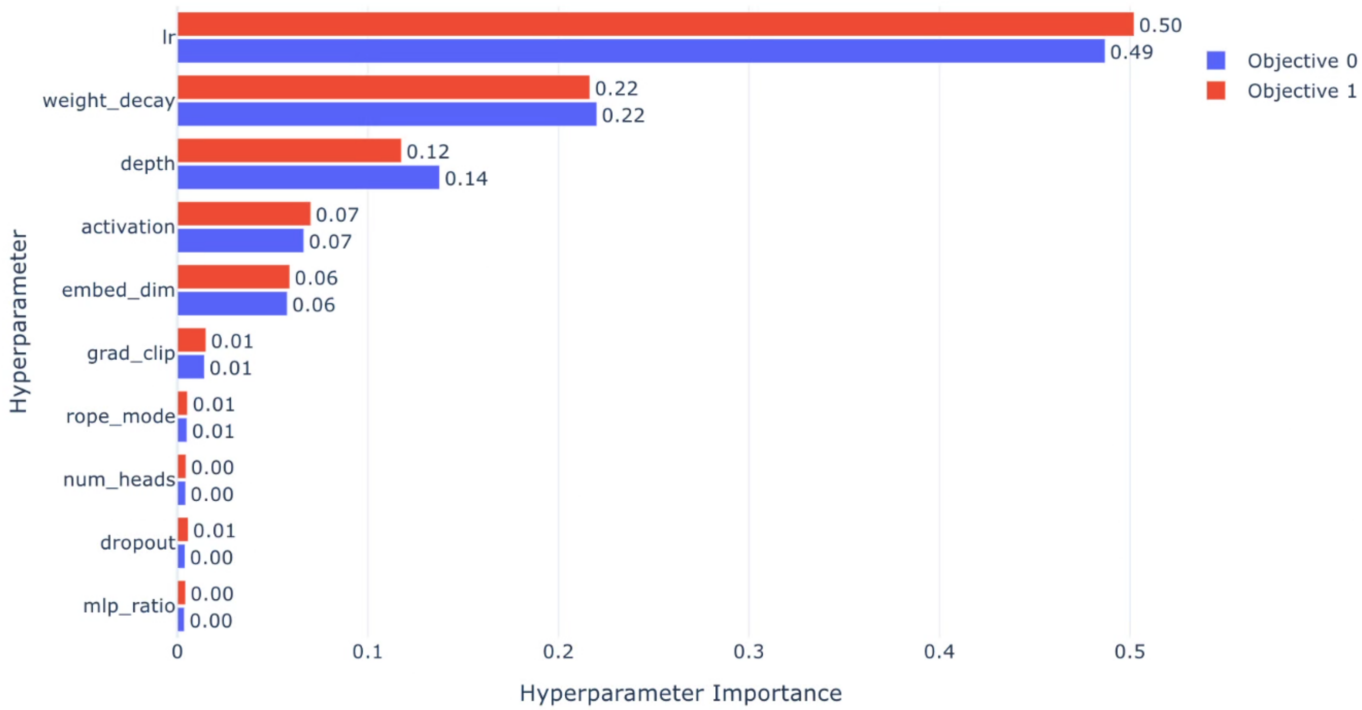



Fig. 2. Hyperparameter importance from the low-fidelity Optuna screening, reported per objective. Objective 0 is the fold-averaged prefix-averaged macro-F1 $\frac{1}{P}\sum_{k=1}^{P}\mathrm{MF1}(k)$ and Objective 1 is the fold-averaged $\mathrm{MF1}(30)$. Importances are normalized to sum to one within each objective, so the values measure relative variance attribution across the 261 completed trials rather than absolute sensitivity.

TABLE V

DISCRETE SEARCH SPACE OF THE LOW-FIDELITY OPTUNA SCREENING ON DREAMS-SUB. EVERY DIMENSION IS CATEGORICAL, SO NO VALUE OUTSIDE THE LISTED GRIDS CAN BE PROPOSED. THE ROTARY BASE IS SAMPLED CONDITIONALLY ON THE ROTARY MODE, AND THE LAST COLUMN REPORTS THE VALUE TAKEN BY THE SELECTED DEFAULT PAUSE CONFIGURATION.

| Group | Hyperparameter | Candidate values | Selected |
|---|---|---|---|
| Architecture | token dimension $d$ | 128, 192, 256 | 128 |
| | Transformer depth | 4, 5, 6, 7, 8, 9, 10, 11, 12 | 7 |
| | attention heads $H$ | 4, 8 | 8 |
| | FFN width ratio | 2, 8/3, 4 | 4 |
| | activation | GELU MLP (classic), SwiGLU | SwiGLU |
| Position | rotary mode | 1D (flattened patch index), 2D (separate patch and epoch subspaces) | 2D |
| | rotary base $\beta$, if 1D | 1000, 1500, 2000, 2500, 3000 | — |
| | rotary base $\beta$, if 2D | 30, 50, 100, 150 | 30 |
| Optimization | peak learning rate | $\{1, 2, \ldots, 9\} \times 10^{-5}$, $\{1, 2, 3, 4, 5\} \times 10^{-4}$ | $3 \times 10^{-4}$ |
| | weight decay | $\{1, 2, \ldots, 9\} \times 10^{-3}$, $\{1, 2, \ldots, 9\} \times 10^{-2}$, $1 \times 10^{-1}$ | $5 \times 10^{-2}$ |
| | dropout rate | 0, 0.1, 0.2, 0.3 | 0.2 |
| | gradient-norm clipping | 1.0, 5.0, $\infty$ (no clipping) | 1.0 |

dimensions; the rotary base is excluded because it is defined only in the subset of trials that selected the matching positional variant, so its importance is not comparable with that of the dimensions sampled in every trial.

The attribution within the 261 completed trials is concentrated in the optimization group. The peak learning rate has the largest relative importance (0.49 for the prefix-averaged objective and 0.50 for MF1(30)), followed by weight decay (0.22 for both objectives). Among the architectural dimensions, Transformer depth has the largest importance (0.14 and 0.12), followed by activation (0.07) and token dimension (0.06). The remaining dimensions have an importance of at most 0.01. These values describe relative attribution within the sampled configurations and search ranges.

The two objectives exhibit similar importance profiles over the completed trials: their rank order is identical, and no dimension differs by more than 0.02. This similarity suggests that, within the explored search space, the same dimensions account for most of the observed variation in both full-prefix and prefix-averaged performance. We retain both objectives during selection and choose the smallest model among the observed Pareto-optimal configurations as shown in Table IV.

### C. Utility-Aware Finite-Horizon Stopper

*1) Default GP Settings:* As described in the main manuscript, the GP-based stopper combines calibrated class posteriors with finite-horizon backward induction to estimate the conditional risk of continuing. For each target fold and acquisition-cost setting $\eta$, we fit $P-1 = 29$ sparse variational GPs, one per nonterminal prefix position, using exclusively out-of-fold predictions from the subjects in the corresponding training split. The GP indexed by $k$ regresses the recursively computed cost-to-go $V_{n,k+1}$ on the $C$-dimensional calibrated posterior $\boldsymbol{\pi}^{\mathrm{cal}}_{n,k}$, and its predictive mean supplies $\mu_k$ in the continuation risk $G^{\mathrm{cont}}_{n,k}$. Its structural settings are listed in Table VI.

The risk that backward induction propagates is the class-weighted cost matrix $\boldsymbol{\Lambda}$ of the main manuscript, for which we set the overall penalty to $\rho = 10$ and the class-balancing exponent to $\alpha = 0.5$. The class counts $M_c$ and the empirical prior $\hat{\boldsymbol{\pi}}$ entering $\boldsymbol{\Lambda}$ are those of the fitting split. Normalizing $w_c$ by $\sum_r \hat{\pi}_r \widetilde{w}_r$ keeps the prior-averaged off-diagonal cost equal to $\rho$ for any $\alpha$, so the acquisition cost $\eta$ retains the same scale when the class weighting changes. We sweep $\eta$ over eight values to trace the speed–accuracy trade-off:

$$\eta \in \begin{Bmatrix} 0.0005,\ 0.006667,\ 0.02,\ 0.03333, \\ 0.06667,\ 0.13333,\ 0.23333,\ 0.5 \end{Bmatrix}$$

Each GP is fitted by maximizing the variational evidence lower bound (ELBO) with Adam, using batches of up to 8192 training states and at most 20 epochs as shown in Table VII. The inducing locations are initialized from the training states and optimized jointly with the remaining GP parameters. Five percent of the training states are held out for validation. After a minimum of five epochs, three consecutive epochs without an improvement of at least $10^{-4}$ in the validation objective trigger one intervention: the best parameters so far are restored and the learning rate is reduced by a factor of 0.25. Because only one such reduction is permitted, a second plateau terminates the fit. We then restore the parameters attaining the best validation objective. This protocol is applied identically across all datasets.

*2) Sensitivity to the Class-Balancing Exponent.:* We use $\alpha = 0.5$ as a moderate class-balancing setting: it corresponds to square-root inverse-frequency weighting and lies between the class-neutral objective ($\alpha = 0$) and full inverse-frequency weighting ($\alpha = 1$). We evaluated $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$ on DREAMS-SUB to decide which value should be chosen. For a fair comparison, Table VIII retains only the three acquisition-cost values shared by all settings. Because the same $\eta$ can induce different acquisition lengths under different class weights,

TABLE VI
SPECIFICATION OF THE UTILITY-AWARE FINITE-HORIZON STOPPER.

| Hyperparameter | Value |
|---|---|
| GP type | sparse variational GP |
| GPs per fold and $\eta$ | $P-1=29$ |
| input statistic | $\boldsymbol{\pi}_{n,k}^{\mathrm{cal}}$ $(C=5)$ |
| regression target | $V_{n,k+1}$ |
| mean function | constant |
| kernel | scaled RBF |
| likelihood | Gaussian |
| inducing points | 400 |
| learn inducing locations | True |
| decision horizon $P$ | 30 patches |
| misclassification penalty $\rho$ | 10.0 |
| class-balancing exponent $\alpha$ | 0.5 |
| acquisition cost $\eta$ | 8 values in $[5\times10^{-4},\,0.5]$ |

TABLE VII
FITTING CONFIGURATION OF THE UTILITY-AWARE FINITE-HORIZON STOPPER.

| Setting | Value |
|---|---|
| objective | variational ELBO |
| optimizer | Adam |
| initial learning rate | 0.02 |
| maximum training epochs | 20 |
| states per batch | 8192 |
| validation fraction | 0.05 |
| early stopping | enabled |
| minimum epochs before stopping | 5 |
| early-stopping patience | 3 epochs |
| minimum improvement | $1\times10^{-4}$ |
| LR reduction factor | 0.25 |
| maximum LR reductions | 1 |
| restored parameters | best validation ELBO |
| random seed | 2025 |

each entry reports both the pooled mean stopping time and macro-F1 values.

Over the common acquisition range, $\alpha = 0.5$ achieves the highest mean interpolated macro-F1 of $79.24\%$, compared with $79.14\%$ for $\alpha = 0.75$, $78.74\%$ for $\alpha = 0.25$, $78.43\%$ for $\alpha = 1$, and $77.15\%$ without class weighting. The small difference between $\alpha = 0.5$ and $0.75$ indicates a broad stable region rather than a sharply identified optimum. We therefore retain $\alpha = 0.5$ as a moderate and robust compromise that improves the macro-F1–acquisition frontier without applying the stronger minority-class reweighting induced by larger exponents.

## V. BASELINE SETTINGS

All comparisons use the same subject-wise 20-fold partition and five-class label mapping. Normalization statistics are fitted on the outer-training fold and reused for its held-out fold. The system-level ECTS baselines are independent acquisition systems with their own classifier and stopping mechanism. The frozen-backbone comparison instead fixes the PAUSE evidence model, class prior, cross-fitted temperatures, folds, and evaluation code, then replaces only the stopping rule. The classic sleep-staging baselines classify a complete 30-s waveform and do not define an adaptive acquisition policy. The tables below report repository defaults and, where these differ, the operating points present in the retained result artifacts.

### A. System-level ECTS baselines

System-level ECTS methods receive the standardized PSD100 sequence $X \in \mathbb{R}^{30\times128}$. Each method may stop after an observed one-second patch and must decide at $k = 30$ if it has not stopped earlier. Because the classifier and stopping mechanism are trained together or fitted as one native pipeline, these experiments compare complete acquisition systems rather than isolated stopping rules.

*a) EARLIEST.:* EARLIEST uses a recurrent state for both classification and stochastic halting [9]. The repository implementation passes the 128-dimensional patch sequence through a two-layer LSTM with hidden width 256. The classifier, stopping controller, and value baseline are implemented as separate two-hidden-layer MLP heads with width 512, supporting sleep-stage prediction, the Bernoulli halting policy, and the policy-gradient baseline. Table IX lists the complete configuration.

*b) ECONOMY-$\gamma$-Max.:* ECONOMY-$\gamma$-Max is a non-myopic cost-based method [12]. It trains an independent XGBoost classifier for each observed prefix, groups posterior trajectories into confidence intervals, and estimates whether stopping or waiting has lower expected cost. The repository calls the pinned `ml_edm` implementation with maximum aggregation and its settings are given in Table X.

*c) EarlyMix.:* EarlyMix is a prefix classifier with sample-level and latent-level mixing, followed by a Transformer decoder that estimates reliability from the posterior trajectory [13]. The classifier is trained on randomly selected prefix lengths, and the decoder learns whether each prefix prediction is correct. At inference, the first reliability score that crosses the swept threshold determines the prediction. Table XI records the architecture and optimization settings.

*d) FIRMBOUND.:* FIRMBOUND casts finite-horizon ECTS as an optimal-stopping problem and solves it through backward induction over the immediate stopping risk and the expected risk of continuation [15]. Tables XII and XIII report the corresponding model and training configurations, respectively.

### B. Frozen-Backbone Decision-Policy Comparison

This comparison asks a narrower question than the system-level experiment: how do different decision policies behave on a shared evidence model? Here, a decision policy includes both the stop-or-continue criterion and the terminal class action taken when stopping; this is the complete mechanism referred to as a stopping rule in the main manuscript. We fix the PAUSE backbone, class prior, cross-fitted temperatures, folds, and evaluation epochs, while each method retains its native

TABLE VIII

SENSITIVITY TO THE CLASS-BALANCING EXPONENT ON DREAMS-SUB. EACH OPERATING POINT REPORTS POOLED $\mathbb{E}[\tau]$/MACRO-F1 (%) OVER 20 LOSO FOLDS. THE FINAL COLUMN IS THE MEAN PIECEWISE-LINEARLY INTERPOLATED MACRO-F1 OVER THE ACQUISITION RANGE SHARED BY ALL SETTINGS, 7.00–16.63 PATCHES.

| $\alpha$ | $\eta = 5 \times 10^{-4}$ $\mathbb{E}[\tau]$/MF1 | $\eta = 2 \times 10^{-2}$ $\mathbb{E}[\tau]$/MF1 | $\eta = 6.67 \times 10^{-2}$ $\mathbb{E}[\tau]$/MF1 | Mean interpolated MF1 |
|---|---|---|---|---|
| 0 | 16.63 / 78.03 | 8.99 / 76.94 | 7.00 / 74.82 | 77.15 |
| 0.25 | 16.69 / 80.00 | 9.37 / 78.41 | 4.73 / 74.30 | 78.74 |
| **0.5** | 17.01 / 80.63 | 9.95 / 79.11 | 4.94 / 75.02 | **79.24** |
| 0.75 | 17.28 / 80.56 | 10.58 / 79.22 | 5.13 / 75.45 | 79.14 |
| 1 | 17.70 / 80.17 | 10.94 / 78.56 | 5.35 / 74.86 | 78.43 |

TABLE IX
EARLIEST SETTINGS.

| Setting | Value |
|---|---|
| input | $30 \times 128$ standardized PSD100 patches |
| recurrent model | two-layer LSTM, hidden width 256 |
| output heads | separate two-hidden-layer MLP heads, width 512, for classification, stopping control, and value baseline |
| class weight | $M/(5M_c)$, computed on the training set |
| optimizer | Adam, learning rate $10^{-3}$ |
| schedule | exponential decay, $\gamma = 0.99$ |
| training budget | 50 epochs, batch size 128 |
| configured sweep | $\lambda \in \{0, 0.001, 0.003, 0.01, 0.03, 0.1, 0.15\}$ |
| retained artifacts | all seven values on DREAMS-SUB and Sleep-EDF-20; $\{0.001, 0.003, 0.01, 0.03, 0.15\}$ on Sleep-EDF-78 and SHHS |
| random seed | 2025 |

TABLE X
ECONOMY-$\gamma$-MAX SETTINGS.

| Setting | Value |
|---|---|
| input | flattened PSD100 prefix at every $k \in \{1, \ldots, 30\}$ |
| prefix classifiers | 30 independent XGBoost models |
| XGBoost | 300 boosting rounds, 5-class softprob output (1500 trees per prefix), depth 6, learning rate 0.05 |
| sampling | row subsample 0.8, column subsample 0.8 |
| tree method | histogram, multiclass soft probabilities |
| trigger | `EconomyGamma`, maximum aggregation |
| configured sweep | $\alpha \in \{0.01, 0.05, 0.1, 0.25, 0.5, 1.0\}$ |
| retained artifacts | configured grid on all datasets, plus $\alpha = 0.001$ on DREAMS-SUB and Sleep-EDF-20 |
| random seed | 2025 |

TABLE XI
EARLYMIX SETTINGS.

| Setting | Value |
|---|---|
| input | $30 \times 128$ standardized PSD100 patches |
| classifier channels | $(64, 128)$, two residual blocks per stage |
| classifier kernel and dropout | kernel 30, dropout 0.3 |
| mixing | sample and latent mixing enabled; four prefix draws per batch |
| reliability decoder | width 256, four heads, two Transformer layers |
| classifier optimization | AdamW, $10^{-3}$, weight decay $10^{-5}$ |
| decoder optimization | AdamW, $10^{-3}$, weight decay $10^{-5}$ |
| training budget | 200 epochs per module; batch size 512 for both modules |
| operating-point sweep | 19 thresholds from 0.05 to 0.95 in steps of 0.05 |
| artifact coverage | all 19 thresholds on all four datasets |
| random seed | 2025 |

TABLE XII
MODEL PARAMETERS OF FIRMBOUND.

| Setting | Value |
|---|---|
| comparison level | independent system-level ECTS pipeline |
| decision horizon | 30 one-second patches |
| evaluation folds | 20 completed folds per dataset |
| stored condition key | acquisition cost $\eta$ |
| SPRT context order | 2 |
| DRE architecture | TANDEMformer (Transformer backbone); input feature dimension 128; embedding dimension 208 |
| Transformer configuration | 3 blocks; 8 attention heads; feed-forward dimension 384; MLP hidden dimension 128 |
| DRE regulation | dropout 0.1 |
| temperature calibration | enabled, range $[0.05, 20]$ |
| backward-induction model | sparse GP regression |
| GP inducing points | 200 |
| continuation-state statistic | posterior probabilities |
| stopping-risk penalty $L$ | 10.0 |

decision mechanism. Fixed, Threshold, GP, and AAGC use calibrated MAP scores and output the corresponding MAP prediction, whereas PAUSE uses the class-weighted utility with $\alpha = 0.5$ and outputs the corresponding class-weighted Bayes action. Each adaptive policy performs its own closed-loop rollout, and its stopping times are written back as endpoint indices so that later epochs receive the history generated by that policy. The experiment therefore isolates the decision layer from representation learning.

*a) Fixed prefix.:* The fixed-prefix rule assigns the same acquisition length $K$ to every epoch. It reads the calibrated MAP prediction at prefix $K$, and all historical endpoints are correspondingly fixed to $\tau_m = K$, so that every preceding epoch contributes the summary $\mathbf{h}_{m,K}$ of its first $K$ patches.

TABLE XIII
TRAINING AND EVALUATION PARAMETERS OF FIRMBOUND.

| Training parameter | Value |
|---|---|
| DRE training setup | 100 epochs; batch size 256; Lion optimizer |
| DRE optimization | learning rate $3\times10^{-4}$; weight decay $10^{-3}$; StepLR scheduler (step size 7, decay factor 0.1) |
| model selection | macro recall (MacRec), class balancing enabled |
| loss weights | multiplet: 0.7; LLLR: 0.1; AUSAT: 0.0 |
| GP training setup | 30 epochs per reverse-time step; batch size 8192; learning rate 0.02 |
| evaluation setup | 20 completed folds per dataset; conditions indexed by acquisition cost $\eta$ |
| internal cost scaling | $c = \eta L/T$, where $L = 10$ and $T = 30$ |
| DREAMS-SUB and Sleep-EDF-20 | $\eta \in \{0.0005, 0.006667, 0.02, 0.03333, 0.06667, 0.13333, 0.23333, 0.5\}$ |
| Sleep-EDF-78 and SHHS | $\eta \in \{0.0005, 0.006667, 0.03333, 0.13333, 0.5\}$ |

Table XIV gives its settings.

TABLE XIV
FIXED-PREFIX SETTINGS ON THE FROZEN PAUSE BACKBONE.

| Setting | Value |
|---|---|
| decision rule | stop every epoch at the same prefix $K$ |
| prediction | calibrated MAP at $K$ |
| history endpoint | $\tau_m = K$ for every preceding epoch |
| sweep | $K \in \{1, \ldots, 30\}$ |
| fitted parameters | none |
| terminal decision | $K = 30$ is the full-horizon point |

*b) Static threshold.:* The static-threshold rule applies the same threshold at every prefix and for every class. It stops when the gap in nats between the two largest calibrated log-posteriors $s_{n,k}(c)/T_k + \log \widehat{\pi}_c$ first reaches $a$. This is the class-independent static boundary used in FIRMBOUND, evaluated here on the calibrated posterior shared by all frozen-backbone rules. Table XV lists the evaluated thresholds.

TABLE XV
STATIC-THRESHOLD SETTINGS ON THE FROZEN PAUSE BACKBONE.

| Setting | Value |
|---|---|
| decision statistic | top-two gap of the calibrated log-posterior (nats) |
| threshold structure | constant across prefixes and classes |
| sweep | $a \in \{0.1, 0.5, 1, 2, 3, 5, 8, 12\}$ |
| prediction | calibrated MAP at the first crossing |
| history handling | rule-specific closed-loop endpoint write-back |
| terminal decision | forced at $k = 30$ |

*c) AAGC.:* AAGC learns a maximum-posterior threshold for each prefix [24]. In our implementation, the model greedily chooses thresholds under a one-sided accumulated gap budget measured against the full-horizon prediction. The settings are shown in Table XVI.

TABLE XVI
AAGC SETTINGS ON THE FROZEN PAUSE BACKBONE.

| Setting | Value |
|---|---|
| decision statistic | maximum calibrated posterior at each prefix |
| threshold structure | one learned threshold per prefix |
| candidate thresholds | $\{0.50, 0.55, 0.60, 0.65, 0.70, 0.75, 0.80, 0.85, 0.90, 0.95, 0.99\}$ |
| gap sweep | $\alpha_{\mathrm{gap}} \in \{0, 0.01, 0.02, 0.03, 0.05, 0.08, 0.15\}$ |
| artifact coverage | seven gap values on DREAMS-SUB and Sleep-EDF-20; $\{0, 0.01, 0.03, 0.05, 0.15\}$ on Sleep-EDF-78 and SHHS |
| terminal decision | final-prefix threshold fixed to zero |

*d) GP.:* The class-neutral GP control retains the finite-horizon backward induction used by PAUSE but removes class-dependent utility weighting. It is the $\alpha = 0$ control on the shared PAUSE backbone; the system-level FIRMBOUND baseline uses its own evidence model. At each nonterminal prefix, a sparse variational GP maps the calibrated five-class posterior to the recursively computed next-step cost-to-go. Its predictive mean estimates continuation risk. The comparison artifacts contain five acquisition costs even though the current GP configuration exposes an eight-cost default grid. Table XVII separates the fitted model settings from the retained sweep.

TABLE XVII
CLASS-NEUTRAL GP SETTINGS ON THE FROZEN PAUSE BACKBONE.

| Setting | Value |
|---|---|
| decision state | calibrated posterior, $C = 5$ |
| GP structure | 29 sparse variational GPs; constant mean, scaled RBF kernel, Gaussian likelihood |
| regression target | recursively computed $V_{k+1}$ |
| misclassification penalty | $\rho = 10$, class exponent $\alpha = 0$ |
| inducing points | 400, learned locations |
| optimization | Adam, learning rate 0.02, variational ELBO |
| fitting budget | at most 20 epochs, batch size 8192 |
| stopping | minimum 5 epochs; patience 3; minimum change $10^{-4}$ |
| learning-rate intervention | factor 0.25, at most once |
| retained cost sweep | $\eta \in \{0.0005, 0.02, 0.06667, 0.13333, 0.5\}$ |
| history handling | rule-specific closed-loop endpoint write-back |

## C. Classic sleep-staging baselines

The classic sleep-staging comparison concerns full-epoch classification, not when to stop acquisition. These models operate on the retained single-channel waveform $X \in \mathbb{R}^{1\times3000}$ and do not use endpoint memory, paired-history training, per-prefix calibration, or adaptive stopping.

*a) AttnSleep.:* AttnSleep combines a multi-resolution CNN with temporal-context encoding based on multi-head attention [25]. The fine branch starts from a 50-sample kernel and resolves transient graphoelements such as spindles and K-complexes, whereas the coarse branch starts from a 400-sample kernel and captures the slow rhythms that separate N3

from the lighter stages. The two branches are concatenated along time, compressed by a residual squeeze-and-excitation block that reweights the merged feature maps, and refined by an encoder whose keys and values come from causal convolutions instead of pointwise linear projections. Table XVIII lists the configuration.

TABLE XVIII
ATTNSLEEP ARCHITECTURE EXPOSED BY THE REPOSITORY.

| Setting | Value |
|---|---|
| input | single-channel 3,000-sample waveform |
| multi-resolution kernels | 50 and 400 samples in the first convolutions |
| CNN dropout | 0.5 |
| temporal-context encoder | two layers |
| model width and FFN | $d_{\text{model}} = 80$, $d_{\text{ff}} = 120$ |
| attention | five heads, dropout 0.1 |
| loss | class-weighted CE loss |
| optimizer | Adam, lr=$10^{-3}$, weight decay=$10^{-3}$ |
| training budget | 100 epochs, batch size 256 |

*b) LGSleepNet.:* LGSleepNet models the waveform through a local CNN branch and a global Transformer branch, then combines them with orthogonal fusion and generalized-mean pooling [26]. Table XIX records settings.

TABLE XIX
LGSLEEPNET SETTINGS.

| Setting | Value |
|---|---|
| global branch | 30 patches of 100 samples; width 100; two Transformer blocks; five heads |
| global-branch dropout | attention and projection dropout 0.2 |
| local branch | convolution channels $64, 64, 128, 256$; CNN dropout 0.5 |
| fusion and pooling | orthogonal fusion and GeM |
| loss | PolyLoss |
| optimizer | Lion, learning rate $5 \times 10^{-5}$, $\beta = (0.9, 0.99)$, weight decay $10^{-3}$ |
| training budget | 100 epochs, batch size 256 |

*c) MSA-CNN.:* MSA-CNN separates multi-scale temporal filtering, spatial filtering, and lightweight temporal attention [27]. Its multi-scale module average-pools the epoch by factors $1, 2, 4, 8$ and applies a kernel of the same length to each scale, so the effective receptive field doubles from one scale to the next at a constant parameter count per scale; a complementary max-pooling by the reciprocal factor returns all scales to a common resolution, and a scale-integration convolution fuses them instead of keeping independent pathways. We use the single-channel `large` variant. Table XX lists the configuration and the learning rates.

*d) ULW-SleepNet.:* ULW-SleepNet uses depthwise-separable residual convolutions and global average pooling to keep the full-epoch classifier small [28]. For the single-channel input, three residual stages increase the feature width before a two-layer classification head. The loss adds the upstream kernel penalty only to the main-path separable convolutions. Table XXI gives the model and training settings.

TABLE XX
MSA-CNN SETTINGS.

| Setting | Value |
|---|---|
| input | $1 \times 1 \times 3000$, adapted from the stored waveform |
| variant | `large`, four temporal scales; input scaling enabled |
| temporal filters | kernel 15, eight channels per scale |
| scale and spatial stages | kernel 5; output widths 32 and 64 |
| attention | embedding width 32, four heads, two layers |
| dropout | 0.1 |
| output and loss | model softmax followed by cross-entropy |
| optimizer | Adam, weight decay $10^{-4}$ |
| learning rate | $2 \times 10^{-4}$ on DREAMS-SUB; $8 \times 10^{-4}$ on Sleep-EDF-20, Sleep-EDF-78, and SHHS |
| training budget | 100 epochs, batch size 512 |
| schedule and early stopping | no scheduler; patience 40 |

TABLE XXI
ULW-SLEEPNET SETTINGS.

| Setting | Value |
|---|---|
| input | $1 \times 3000$ waveform |
| residual stages | three stages with filters $(8, 16, 32)$ |
| separable convolution | kernel 3; max pooling by 2 in each light CNN |
| dropout | 0.1 after residual stages, 0.3 before the head |
| classification head | $32 \to 64 \to 5$ |
| loss | cross-entropy plus main-path kernel L2, coefficient $10^{-3}$ |
| optimizer | Adam with AMSGrad, learning rate $10^{-3}$, no optimizer weight decay |
| schedule | cosine warm restarts per update, $T_0 = 10$, $T_{\text{mult}} = 2$ |
| training budget | 100 epochs, batch size 1024 |
| early stopping and selection | patience 40 |

## VI. COMPLETE ANALYSIS ON MAIN EXPERIMENTS

### A. Class-Wise Speed–Accuracy Analysis

As shown in Fig. 3, the class-wise frontiers show that PAUSE's system-level advantage is concentrated in N1 and REM rather than being driven uniformly by all five stages. As summarized in Table XXII, the average matched-budget gain is 13.30–19.50 F1 points for N1 and 13.20–20.36 points for REM. These are also the two largest gains on every dataset, averaging 16.15 and 17.35 points across datasets, respectively. At the peak of each method's class-wise frontier, PAUSE exceeds the strongest baseline by 10.41–21.96 points on N1 and by 11.53–15.11 points on REM. Moreover, PAUSE reaches or exceeds the best baseline peak after only 3.9–6.3 N1 patches and 2.1–4.4 REM patches on average for the corresponding reference stage; the baseline peaks require 25.0–29.9 and 25.3–29.9 patches, respectively. Thus, aggressive acquisition reduction does not obtain its macro-F1 advantage by abandoning the difficult N1 stage. The similarly large REM separation further shows that the benefit is not confined to the single minority class emphasized in the main paper.

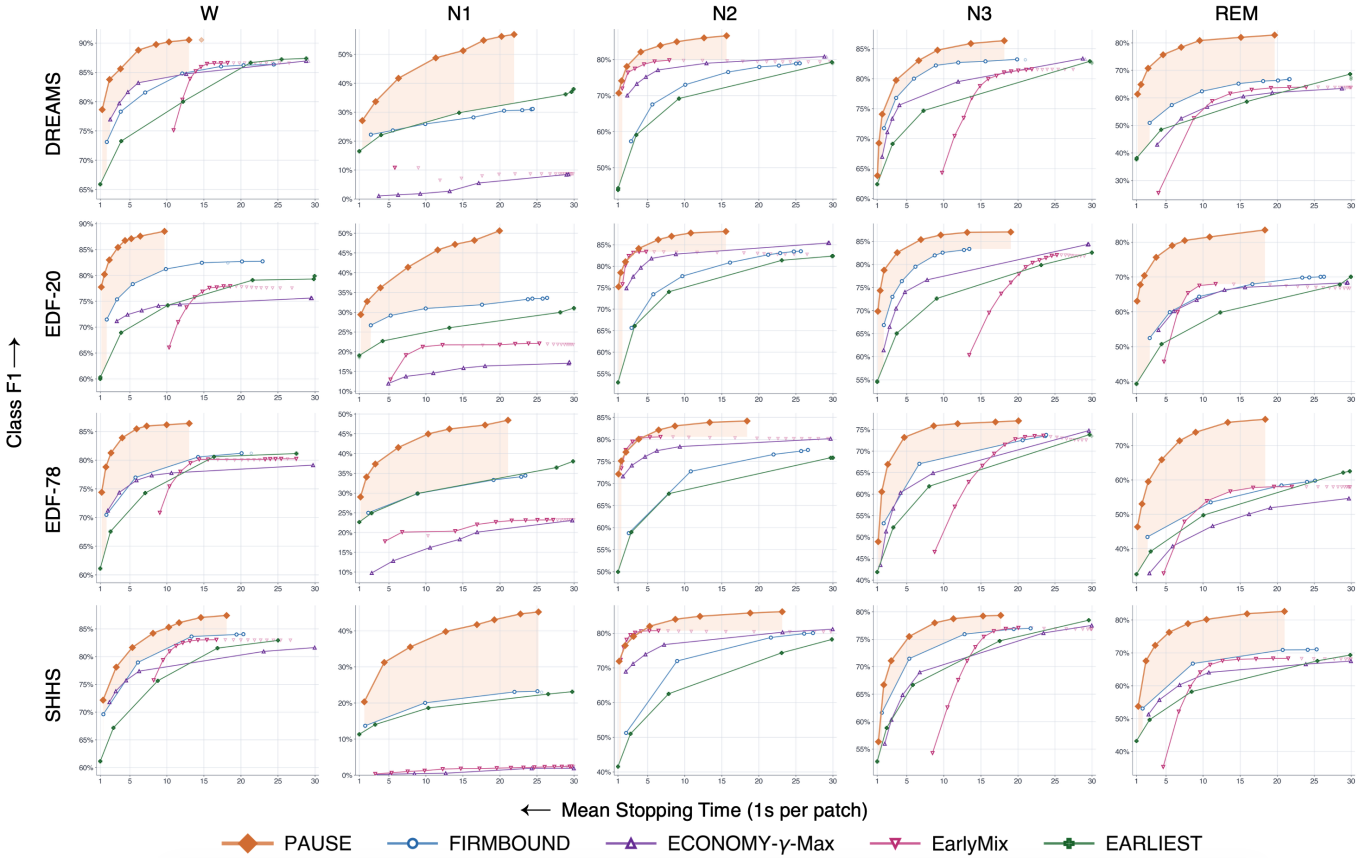



Fig. 3. Class-wise speed–accuracy frontiers for the system-level ECTS comparison. Rows correspond to datasets and columns to sleep stages. The horizontal axis is the reference-stage-conditional mean stopping time $\mathbb{E}[\tau \mid c]$, and the vertical axis is pooled class-wise F1. Solid lines connect the within-method Pareto fronts, whereas faint markers denote evaluated but dominated operating points. Orange shading marks budgets at which PAUSE exceeds the upper envelope of the four baselines; grey shading marks the converse. All values are reconstructed from the retained 20-fold pooled per-class summaries.

TABLE XXII
AVERAGE MATCHED-BUDGET PARETO GAIN OF PAUSE OVER THE JOINT UPPER ENVELOPE OF EARLIEST, ECONOMY-$\gamma$-MAX, EARLYMIX, AND FIRMBOUND, IN CLASS-WISE F1 PERCENTAGE POINTS. THE VALUES USE THE MATCHED-BUDGET INTEGRAL DEFINED IN THE EVALUATION METRICS; BOLD IDENTIFIES THE TWO LARGEST STAGE-SPECIFIC GAINS ON EACH DATASET.

| Dataset | W | N1 | N2 | N3 | REM |
|---|---|---|---|---|---|
| DREAMS-SUB | 6.29 | **19.50** | 5.56 | 3.01 | **18.86** |
| Sleep-EDF-20 | 9.70 | **13.30** | 3.79 | 5.43 | **16.96** |
| Sleep-EDF-78 | 8.33 | **13.55** | 2.60 | 7.92 | **20.36** |
| SHHS | 3.65 | **18.25** | 3.61 | 3.92 | **13.20** |
| Mean | 6.99 | **16.15** | 3.89 | 5.07 | **17.35** |

The remaining stages exhibit smaller but mostly consistent improvements. For W and N3, the average matched-budget gains are 3.65–9.70 and 3.01–7.92 points, respectively, and PAUSE remains above the baseline envelope throughout every comparable interval. N2 has the smallest average gain (2.60–5.56 points) and is the only stage for which the advantage is not uniform over budget. The minimum N2 gap is $-0.56$, $-0.87$, and $-1.39$ points on Sleep-EDF-20, Sleep-EDF-78, and SHHS, although the gap is nonnegative over 93.3%, 88.5%, and 88.0% of their comparable budget intervals and remains positive throughout DREAMS-SUB. PAUSE still attains a higher peak N2 F1 than the strongest baseline by 2.64–5.76 points. The evidence therefore supports a precise interpretation: the largest and most consistent class-wise improvements occur for N1 and REM, W and N3 provide moderate supporting gains, and N2 shows a smaller frontier-wide advantage with narrow baseline-leading regions on three datasets.

Overall, the class-wise analysis shows that PAUSE improves the system-level speed–accuracy trade-off without relying on a single favorable stage or systematically sacrificing any class. The dominant gains on N1 and REM account for most of the macro-F1 improvement, while the consistent benefits on W and N3 and the predominantly positive N2 gaps demonstrate that the advantage extends across the full label space. Although several narrow N2 budget intervals favor the baselines, PAUSE still achieves a higher peak F1 for every stage and remains superior over most comparable budgets. These results indicate that PAUSE reduces acquisition substantially while preserving, and in most cases improving, class-wise discrimination, with particularly strong benefits for the more challenging N1 and REM stages.

*a) Fold-level results at low acquisition budgets.:* Table XXIII reports the mean and sample standard deviation of macro-F1, N1 F1, and acquisition length over 20 test folds at two low-budget PAUSE settings, $\eta = 0.5$ and $0.23333$. Each baseline is compared at the operating point closest to PAUSE in pooled mean acquisition length.

PAUSE still achieves higher mean macro-F1 and N1 F1 than all four ECTS baselines at both settings. At $\eta = 0.5$, it acquires only $1.103$–$1.223$ patches per epoch across the four datasets, exceeding the best baseline mean by $2.06$–$6.22$ macro-F1 points and $2.02$–$5.40$ N1 F1 points. At $\eta = 0.23333$, it acquires $1.481$–$2.269$ patches, with larger gains of $9.15$–$14.09$ macro-F1 points and $5.32$–$16.93$ N1 F1 points.

### B. Stopper Speed-Accuracy Analysis

Since all five rules use the same backbone, calibration procedure, folds, and evaluation epochs, to generate their own closed-loop histories, this experiment isolates the decision layer from representation learning more directly than the system-level ECTS comparison. Because every policy retains its native stopping class action, and the comparison evaluates complete decision policies rather than stopping times alone. As shown in Fig. 4, PAUSE stays above the alternative-rule envelope over the entire common macro-F1 budget interval on every dataset. Its average matched-budget gains are $2.21$, $1.78$, $3.55$, and $3.92$ macro-F1 points on DREAMS-SUB, Sleep-EDF-20, Sleep-EDF-78, and SHHS, respectively. The advantage also holds at the frontier endpoints: PAUSE reaches peak macro-F1 values of $80.63\%$, $79.56\%$, $74.74\%$, and $76.16\%$, exceeding the best alternative peaks by $0.71$, $0.89$, $1.72$, and $2.68$ points while using $29.1\%$–$46.6\%$ fewer patches. The improvement is therefore not restricted to a single acquisition-cost setting.

The class-wise decomposition further identifies where the advantage of the complete PAUSE decision policy is concentrated. N1 improves most consistently: its average matched-budget gap is positive throughout the comparable interval on all datasets and ranges from $3.34$ points on DREAMS-SUB to $15.11$ points on SHHS. PAUSE also raises the peak N1 F1 by $2.83$–$12.22$ points over the strongest alternative peak. N3 provides the second major contribution, with average gains of $1.20$, $2.93$, $9.84$, and $5.28$ points. The particularly large Sleep-EDF-78 N3 gap is frontier-wide rather than a peak-only effect: PAUSE reaches $77.04\%$ peak N3 F1, $4.85$ points above the strongest alternative, and already exceeds that alternative's peak after an average of $4.7$ N3 patches. REM exhibits a smaller but positive matched-budget gain of $0.70$–$1.89$ points across the four datasets.

It can also be found that the gain is not uniform across stages. W and N2 have small negative average gaps on every dataset, ranging from $-0.26$ to $-0.16$ points for W and from $-0.45$ to $-0.27$ points for N2. Their peak deficits remain below $0.95$ and $0.46$ points, respectively. REM likewise has a marginally lower unconstrained peak ($0.02$–$0.85$ points) despite its positive matched-budget gap, indicating an earlier-decision advantage rather than a higher full-budget ceiling. This controlled experiment clarifies the stopper-side roles of the two design principles in PAUSE. Prefix adaptivity turns accumulating evidence into an epoch-specific acquisition length rather than imposing one fixed observation budget, whereas utility awareness reshapes that decision under a class-balanced objective. Empirically, the latter is expressed mainly through the large N1 and N3 gains that outweigh the small W and N2 reductions. The positive matched-budget macro-F1 gap therefore supports the effectiveness of PAUSE's complete utility-aware stopping policy on a fixed evidence model.

*a) Fold-level comparison of low-budget stopping policies.:* Table XXV compares the low-budget stopping policies on the frozen PAUSE backbone. PAUSE achieves the highest mean N1 F1 in all eight dataset–cost settings, exceeding the best alternative mean by $0.57$–$18.29$ percentage points. On Sleep-EDF-20 at $\eta = 0.23333$, PAUSE and the nearest GP operating point reach similar mean macro-F1 ($65.33\%$ versus $65.21\%$), while PAUSE improves mean N1 F1 by $2.57$ points and uses $21.4\%$ fewer patches ($1.481$ versus $1.884$). On SHHS at the same PAUSE cost, it improves mean macro-F1 by $3.61$ points and mean N1 F1 by $18.29$ points over the nearest GP point, with $21.2\%$ fewer patches.

### C. Detailed Ablation Study

The ablation of the main manuscript reports macro-F1 against the mean acquisition length and establishes that both components contribute to the frontier. Fig. 5 resolves the same eight-cost sweep by stage, with each variant performing its own closed-loop rollout so that the endpoint history visible to later epochs is the one its own policy generates. Table XXVI reports the matched-budget Pareto gains of PAUSE over each ablated variant, using the integral defined in the evaluation-metrics section. Against the joint envelope of the two ablations, PAUSE holds a positive macro-F1 gap of $6.58$, $3.47$, $2.31$, and $1.97$ points over the entire comparable budget interval on DREAMS-SUB, Sleep-EDF-20, Sleep-EDF-78, and SHHS, and attains the peak macro-F1 of the stronger ablation using $13.5\%$–$52.6\%$ fewer patches.

Removing endpoint memory produces a stage profile concentrated in N1 and REM, whose average matched-budget deficits are $10.80$–$20.09$ and $10.12$–$16.79$ points against $1.82$–$8.64$ points for W, N2, and N3. The attainable peaks separate in the same way, by $10.34$–$22.15$ points for N1 and $7.51$–$11.53$ for REM. These are also the two stages that prior cross-epoch staging work identifies as most dependent on surrounding epochs [16], [17], so the benefit of endpoint memory falls where an epoch is least self-sufficient. N3 is the clear exception, and its gap is budget-dependent rather than uniform: on DREAMS-SUB it falls from $13.82$ points at the cheapest comparable budget to $0.12$ points at the most expensive, and the memory-free variant eventually reaches $86.12\%$ peak N3 F1 against $86.35\%$ for PAUSE. Endpoint memory therefore accelerates the recognition of N3 rather than raising its ceiling.

Removing paired-history training yields a smaller and differently shaped deficit, with a cross-dataset mean of $3.73$

TABLE XXIII

LOW-BUDGET SYSTEM-LEVEL COMPARISON. ENTRIES ARE MEAN ± SAMPLE STANDARD DEVIATION OVER 20 FOLDS. $\Delta b$ IS BASELINE MINUS PAUSE POOLED MEAN PATCHES; PAIRED VALUES FOLLOW THE COST ORDER. THE SECOND PAUSE COST IS ABBREVIATED AS 0.233.

| **Dataset** | **PAUSE** $\eta$ | **Method (setting)** | **Macro-F1** (%) | **N1 F1** (%) | **Patches** | $\Delta b$ |
|---|---|---|---|---|---|---|
| DREAMS-SUB | 0.5 | **PAUSE** | $57.04 \pm 6.03$ | $24.01 \pm 9.53$ | $1.154 \pm 0.049$ | 0 |
| | 0.233 | **PAUSE** | $62.46 \pm 6.15$ | $32.26 \pm 12.17$ | $1.718 \pm 0.190$ | 0 |
| | 0.5 / 0.233 | EARLIEST ($\lambda = 0.1$) | $43.73 \pm 3.73$ | $15.76 \pm 5.71$ | $1.023 \pm 0.024$ | $-0.130/-0.693$ |
| | 0.5 / 0.233 | ECONOMY-$\gamma$-Max ($\alpha = 1$) | $49.45 \pm 4.78$ | $1.05 \pm 2.02$ | $2.482 \pm 0.418$ | $+1.332/+0.769$ |
| | 0.5 / 0.233 | EarlyMix ($\theta = 0.05$) | $45.45 \pm 8.42$ | $9.80 \pm 8.20$ | $5.501 \pm 1.395$ | $+4.317/+3.753$ |
| | 0.5 / 0.233 | FIRMBOUND ($\eta = 0.5$) | $53.13 \pm 4.79$ | $20.78 \pm 7.18$ | $2.424 \pm 0.405$ | $+1.275/+0.712$ |
| Sleep-EDF-20 | 0.5 | **PAUSE** | $61.55 \pm 5.28$ | $27.16 \pm 9.17$ | $1.103 \pm 0.038$ | 0 |
| | 0.233 | **PAUSE** | $65.33 \pm 4.44$ | $30.46 \pm 10.48$ | $1.481 \pm 0.130$ | 0 |
| | 0.5 / 0.233 | EARLIEST ($\lambda = 0.1$) | $43.66 \pm 4.22$ | $17.00 \pm 6.59$ | $1.010 \pm 0.021$ | $-0.093/-0.469$ |
| | 0.5 / 0.233 | ECONOMY-$\gamma$-Max ($\alpha = 1$) | $53.21 \pm 5.50$ | $9.33 \pm 10.07$ | $2.808 \pm 0.400$ | $+1.710/+1.333$ |
| | 0.5 / 0.233 | EarlyMix ($\theta = 0.05$) | $49.61 \pm 9.05$ | $11.24 \pm 9.00$ | $5.652 \pm 1.713$ | $+4.409/+4.033$ |
| | 0.5 / 0.233 | FIRMBOUND ($\eta = 0.5$) | $55.33 \pm 5.98$ | $25.14 \pm 9.85$ | $2.514 \pm 0.433$ | $+1.413/+1.037$ |
| Sleep-EDF-78 | 0.5 | **PAUSE** | $51.51 \pm 6.80$ | $27.36 \pm 7.14$ | $1.125 \pm 0.065$ | 0 |
| | 0.233 | **PAUSE** | $58.60 \pm 5.51$ | $33.06 \pm 5.20$ | $1.640 \pm 0.210$ | 0 |
| | 0.5 / 0.233 | EARLIEST ($\lambda = 0.15$) | $40.84 \pm 3.79$ | $21.90 \pm 3.77$ | $1.001 \pm 0.002$ | $-0.125/-0.643$ |
| | 0.5 / 0.233 | ECONOMY-$\gamma$-Max ($\alpha = 1$) | $45.18 \pm 3.86$ | $10.17 \pm 3.86$ | $2.020 \pm 0.216$ | $+0.902/+0.384$ |
| | 0.5 / 0.233 | EarlyMix ($\theta = 0.05$) | $46.57 \pm 6.54$ | $16.57 \pm 7.98$ | $4.576 \pm 0.694$ | $+3.466/+2.948$ |
| | 0.5 / 0.233 | FIRMBOUND ($\eta = 0.5$) | $49.45 \pm 5.30$ | $24.48 \pm 4.59$ | $2.252 \pm 0.389$ | $+1.119/+0.601$ |
| SHHS | 0.5 | **PAUSE** | $54.30 \pm 3.91$ | $19.13 \pm 6.65$ | $1.223 \pm 0.057$ | 0 |
| | 0.233 | **PAUSE** | $63.91 \pm 2.06$ | $30.94 \pm 4.41$ | $2.269 \pm 0.155$ | 0 |
| | 0.5 | EARLIEST ($\lambda = 0.15$) | $41.97 \pm 2.01$ | $11.35 \pm 1.30$ | $1.000 \pm 0.000$ | $-0.224$ |
| | 0.233 | EARLIEST ($\lambda = 0.03$) | $48.13 \pm 2.25$ | $14.01 \pm 1.96$ | $2.663 \pm 0.132$ | $+0.393$ |
| | 0.5 / 0.233 | ECONOMY-$\gamma$-Max ($\alpha = 1$) | $49.63 \pm 1.83$ | $0.13 \pm 0.24$ | $2.239 \pm 0.195$ | $+1.014/-0.033$ |
| | 0.5 / 0.233 | EarlyMix ($\theta = 0.05$) | $47.41 \pm 3.02$ | $0.35 \pm 0.76$ | $4.301 \pm 0.453$ | $+3.076/+2.030$ |
| | 0.5 / 0.233 | FIRMBOUND ($\eta = 0.5$) | $49.82 \pm 2.16$ | $13.73 \pm 1.57$ | $1.849 \pm 0.298$ | $+0.625/-0.421$ |

TABLE XXIV

AVERAGE MATCHED-BUDGET PARETO GAP BETWEEN PAUSE AND THE JOINT UPPER ENVELOPE OF FIXED, THRESHOLD, AAGC, AND GP IN F1 PERCENTAGE POINTS. THE MACRO-F1 COLUMN USES THE OVERALL SUPPORT-WEIGHTED MEAN ACQUISITION LENGTH; EACH STAGE COLUMN USES $\mathbb{E}[\tau \mid c]$. POSITIVE VALUES FAVOR PAUSE. BOLD MARKS THE LARGEST STAGE-SPECIFIC GAIN ON EACH DATASET.

| **Dataset** | **Macro-F1** | **W** | **N1** | **N2** | **N3** | **REM** |
|---|---|---|---|---|---|---|
| DREAMS-SUB | 2.21 | -0.16 | **3.34** | -0.45 | 1.20 | 1.71 |
| Sleep-EDF-20 | 1.78 | -0.22 | **3.57** | -0.45 | 2.93 | 0.70 |
| Sleep-EDF-78 | 3.55 | -0.26 | 6.20 | -0.27 | **9.84** | 1.89 |
| SHHS | 3.92 | -0.18 | **15.11** | -0.37 | 5.28 | 1.44 |
| Mean | 2.87 | -0.21 | **7.06** | -0.38 | 4.81 | 1.43 |

macro-F1 points against 8.70 for the memory-free variant. The two ablations separate most clearly on N3, for which paired-history training accounts for 3.80 points on average while endpoint memory accounts for 3.44, reversing their relative order on every other stage. The components therefore do not act through a single shared mechanism, which complements the budget-axis dissociation reported in the main manuscript. The advantage is also not uniform over every budget: on Sleep-EDF-78 the N1 gap reaches $-1.16$ points at the cheapest comparable budgets, the only negative region among the 24 dataset and stage combinations, although it stays nonnegative over 95.7% of that interval.

### *D. Detailed Full-Observation Sleep-Staging Comparison*

The system-level ECTS results above jointly reflect representation learning and adaptive stopping. Table XXVII instead isolates the classification capability of the backbone by reporting its complete sleep-staging performance under full observation. This evaluation is important because PAUSE is ultimately designed for practical sleep staging, where reliable early stopping requires a strong classifier that can provide sufficiently discriminative evidence from partial observations. Otherwise, the stopping policy would be forced to act on weak or ambiguous predictions. The full-observation experiment therefore verifies that PAUSE is built upon a competitive classification foundation rather than obtaining efficiency gains by sacrificing backbone accuracy. The table expands the compact overall results in the main paper with five class-wise F1 scores; all entries use equal-weight fold means. The results can show classification capability rather than acquisition efficiency, as it contains neither adaptive stopping times nor matched-budget

TABLE XXV

LOW-BUDGET STOPPING-RULE COMPARISON ON THE SHARED PAUSE BACKBONE. ENTRIES ARE MEAN ± SAMPLE STANDARD DEVIATION OVER 20 FOLDS. $\Delta b$ IS BASELINE MINUS PAUSE POOLED MEAN PATCHES; PAIRED VALUES FOLLOW THE COST ORDER. THE SECOND PAUSE COST IS ABBREVIATED AS 0.233.

| Dataset | PAUSE $\eta$ | Method (setting) | Macro-F1 (%) | N1 F1 (%) | Patches | $\Delta b$ |
|---|---|---|---|---|---|---|
| DREAMS-SUB | 0.5 | **PAUSE** | $57.04 \pm 6.03$ | $24.01 \pm 9.53$ | $1.154 \pm 0.049$ | 0 |
| | 0.233 | **PAUSE** | $62.46 \pm 6.15$ | $32.26 \pm 12.17$ | $1.718 \pm 0.190$ | 0 |
| | 0.5 | Fixed ($k = 1$) | $51.52 \pm 6.79$ | $12.18 \pm 9.89$ | $1.000 \pm 0.000$ | $-0.152$ |
| | 0.233 | Fixed ($k = 2$) | $59.22 \pm 6.88$ | $21.35 \pm 12.70$ | $2.000 \pm 0.000$ | $+0.285$ |
| | 0.5 | Threshold ($\theta = 0.1$) | $52.91 \pm 7.15$ | $13.48 \pm 10.46$ | $1.056 \pm 0.020$ | $-0.097$ |
| | 0.233 | Threshold ($\theta = 0.5$) | $58.69 \pm 7.02$ | $18.99 \pm 12.25$ | $1.605 \pm 0.211$ | $-0.111$ |
| | 0.5 / 0.233 | AAGC ($\alpha_{\text{gap}} = 0.15$) | $56.34 \pm 7.12$ | $16.91 \pm 12.64$ | $1.472 \pm 0.228$ | $+0.318/-0.245$ |
| | 0.5 | GP ($\eta = 0.5$) | $55.18 \pm 6.53$ | $16.22 \pm 12.30$ | $1.185 \pm 0.077$ | $+0.031$ |
| | 0.233 | GP ($\eta = 0.23333$) | $61.34 \pm 5.67$ | $21.83 \pm 12.41$ | $1.805 \pm 0.202$ | $+0.090$ |
| Sleep-EDF-20 | 0.5 | **PAUSE** | $61.55 \pm 5.28$ | $27.16 \pm 9.17$ | $1.103 \pm 0.038$ | 0 |
| | 0.233 | **PAUSE** | $65.33 \pm 4.44$ | $30.46 \pm 10.48$ | $1.481 \pm 0.130$ | 0 |
| | 0.5 / 0.233 | Fixed ($k = 1$) | $56.63 \pm 6.27$ | $20.75 \pm 9.45$ | $1.000 \pm 0.000$ | $-0.102/-0.479$ |
| | 0.5 | Threshold ($\theta = 0.1$) | $57.42 \pm 6.09$ | $20.97 \pm 10.09$ | $1.048 \pm 0.009$ | $-0.055$ |
| | 0.233 | Threshold ($\theta = 0.5$) | $62.17 \pm 5.39$ | $23.57 \pm 10.69$ | $1.552 \pm 0.094$ | $+0.075$ |
| | 0.5 / 0.233 | AAGC ($\alpha_{\text{gap}} = 0.08$) | $59.78 \pm 6.30$ | $24.89 \pm 9.53$ | $1.405 \pm 0.153$ | $+0.303/-0.073$ |
| | 0.5 | GP ($\eta = 0.5$) | $57.93 \pm 6.11$ | $21.47 \pm 10.48$ | $1.076 \pm 0.031$ | $-0.028$ |
| | 0.233 | GP ($\eta = 0.13333$) | $65.21 \pm 4.64$ | $27.89 \pm 10.60$ | $1.884 \pm 0.231$ | $+0.397$ |
| Sleep-EDF-78 | 0.5 | **PAUSE** | $51.51 \pm 6.80$ | $27.36 \pm 7.14$ | $1.125 \pm 0.065$ | 0 |
| | 0.233 | **PAUSE** | $58.60 \pm 5.51$ | $33.06 \pm 5.20$ | $1.640 \pm 0.210$ | 0 |
| | 0.5 | Fixed ($k = 1$) | $36.54 \pm 8.02$ | $11.63 \pm 9.35$ | $1.000 \pm 0.000$ | $-0.125$ |
| | 0.233 | Fixed ($k = 2$) | $46.65 \pm 7.69$ | $19.12 \pm 8.33$ | $2.000 \pm 0.000$ | $+0.357$ |
| | 0.5 | Threshold ($\theta = 0.1$) | $39.55 \pm 7.86$ | $13.75 \pm 9.56$ | $1.085 \pm 0.030$ | $-0.040$ |
| | 0.233 | Threshold ($\theta = 0.5$) | $51.06 \pm 7.45$ | $24.46 \pm 7.12$ | $2.150 \pm 0.338$ | $+0.513$ |
| | 0.5 / 0.233 | AAGC ($\alpha_{\text{gap}} = 0.15$) | $51.65 \pm 6.21$ | $26.79 \pm 5.76$ | $2.530 \pm 0.451$ | $+1.409/+0.891$ |
| | 0.5 | GP ($\eta = 0.5$) | $39.58 \pm 7.37$ | $13.56 \pm 9.12$ | $1.093 \pm 0.044$ | $-0.031$ |
| | 0.233 | GP ($\eta = 0.13333$) | $54.50 \pm 6.13$ | $27.67 \pm 5.42$ | $2.078 \pm 0.291$ | $+0.439$ |
| SHHS | 0.5 | **PAUSE** | $54.30 \pm 3.91$ | $19.13 \pm 6.65$ | $1.223 \pm 0.057$ | 0 |
| | 0.233 | **PAUSE** | $63.91 \pm 2.06$ | $30.94 \pm 4.41$ | $2.269 \pm 0.155$ | 0 |
| | 0.5 | Fixed ($k = 1$) | $33.99 \pm 4.39$ | $0.02 \pm 0.07$ | $1.000 \pm 0.000$ | $-0.224$ |
| | 0.233 | Fixed ($k = 2$) | $46.40 \pm 3.36$ | $0.14 \pm 0.25$ | $2.000 \pm 0.000$ | $-0.270$ |
| | 0.5 | Threshold ($\theta = 0.1$) | $38.92 \pm 4.54$ | $0.03 \pm 0.15$ | $1.085 \pm 0.022$ | $-0.139$ |
| | 0.233 | Threshold ($\theta = 0.5$) | $53.44 \pm 2.81$ | $5.48 \pm 4.33$ | $1.912 \pm 0.133$ | $-0.359$ |
| | 0.5 / 0.233 | AAGC ($\alpha_{\text{gap}} = 0.15$) | $52.91 \pm 3.67$ | $11.48 \pm 5.57$ | $1.967 \pm 0.204$ | $+0.742/-0.304$ |
| | 0.5 | GP ($\eta = 0.5$) | $38.92 \pm 4.53$ | $0.06 \pm 0.28$ | $1.106 \pm 0.029$ | $-0.118$ |
| | 0.233 | GP ($\eta = 0.13333$) | $60.30 \pm 2.06$ | $12.65 \pm 5.09$ | $2.879 \pm 0.210$ | $+0.608$ |

TABLE XXVI

AVERAGE MATCHED-BUDGET PARETO GAIN OF PAUSE OVER EACH ABLATED VARIANT IN F1 PERCENTAGE POINTS. BOLD MARKS THE LARGEST STAGE-SPECIFIC GAIN IN EACH ROW.

| Removed | Dataset | MF1 | W | N1 | N2 | N3 | REM |
|---|---|---|---|---|---|---|---|
| Endpoint Memory | DREAMS-SUB | 9.53 | 4.11 | **19.57** | 6.36 | 1.82 | **15.27** |
| | Sleep-EDF-20 | 7.72 | 8.64 | **10.80** | 3.64 | 4.16 | **11.61** |
| | Sleep-EDF-78 | 8.55 | 5.60 | **14.82** | 3.31 | 3.81 | **16.79** |
| | SHHS | 9.00 | 3.04 | **20.09** | 5.77 | 3.98 | **10.12** |
| | Mean | 8.70 | 5.35 | **16.32** | 4.77 | 3.44 | **13.45** |
| Paired-History | DREAMS-SUB | 7.13 | 3.09 | **15.97** | 2.85 | 6.66 | 7.44 |
| | Sleep-EDF-20 | 3.47 | 2.58 | 3.83 | 2.17 | **5.31** | 3.94 |
| | Sleep-EDF-78 | 2.32 | 1.41 | 2.82 | 1.55 | 2.23 | **3.90** |
| | SHHS | 2.01 | 0.72 | **6.60** | 1.16 | 0.99 | 1.32 |
| | Mean | 3.73 | 1.95 | **7.31** | 1.93 | 3.80 | 4.15 |

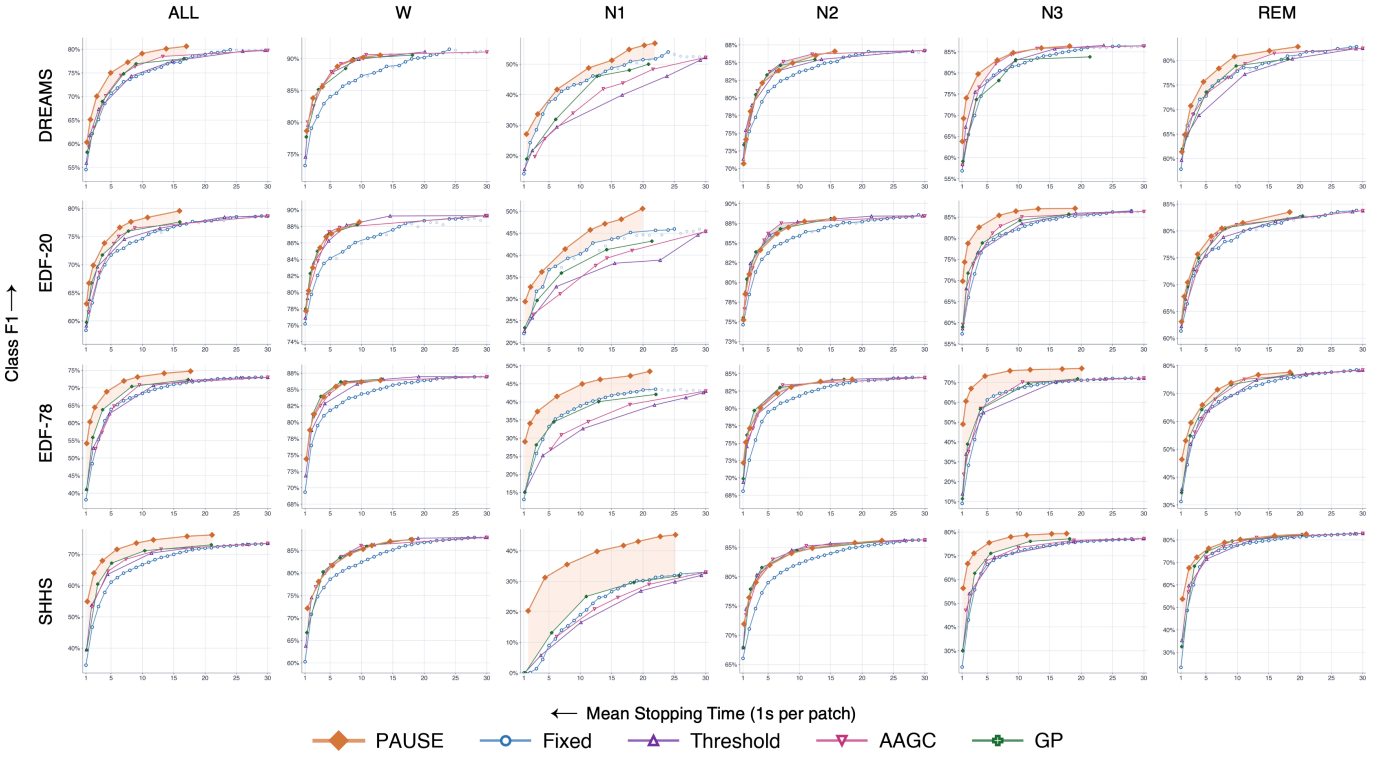



Fig. 4. Class-wise speed–accuracy frontiers of the stopping rules on the frozen PAUSE backbone. Rows correspond to datasets and columns to sleep stages. The horizontal axis is the reference-stage-conditional mean stopping time $\mathbb{E}[\tau \mid c]$, and the vertical axis is pooled class-wise F1. Fixed, Threshold, AAGC, and the class-neutral GP ($\alpha = 0$) use the same calibrated PAUSE evidence. Every adaptive rule performs its own closed-loop rollout. Solid lines join within-rule Pareto points, faint markers denote dominated operating points, and orange shading marks budgets at which PAUSE exceeds the joint upper envelope of the four alternative rules.

operating points.

PAUSE obtains the highest mean ACC on all four datasets and the highest mean macro-F1 on DREAMS-SUB, Sleep-EDF-20, and SHHS, with margins of 3.71, 0.93, and 2.68 percentage points over the strongest comparator, respectively. On Sleep-EDF-78, its mean macro-F1 is 73.52%, compared with 73.90% for AttnSleep and 73.58% for LGSleepNet. For MGm, PAUSE achieves the highest mean on DREAMS-SUB and SHHS, while remaining 0.71 and 1.20 points below the best comparator on Sleep-EDF-20 and Sleep-EDF-78, respectively. The fold standard deviations in the main paper describe substantial between-fold variation, so these mean differences alone do not establish statistical significance. The overall evidence supports a competitive full-observation classifier across the four datasets.

The per-stage fold means clarify where this performance comes from. PAUSE attains the highest N1 and REM F1 on all four datasets. Its N1 margins over the corresponding runner-up are 12.67, 3.26, 2.47, and 9.66 percentage points on DREAMS-SUB, Sleep-EDF-20, Sleep-EDF-78, and SHHS, respectively; its REM margins range from 3.31 to 4.51 points. PAUSE has the highest W F1 on three datasets and trails ULW-SleepNet by 0.93 points on DREAMS-SUB. The N2 results are closer: PAUSE leads on DREAMS-SUB and remains within 0.44–0.52 points of the best comparator on the other three datasets. N3 is the clear weakness, with deficits of 2.95, 4.23, 8.91, and 3.91 points on the four datasets, respectively. Hence, the backbone's macro-F1 strength is associated primarily with N1 and REM, whereas its N3 representation leaves room for improvement.

Taken together, these analyses separate both the sources and the limits of PAUSE's system-level improvement. In the system-level comparison, all dataset and stage combinations show positive average matched-budget gains, with the largest cross-dataset improvements occurring for N1 (16.15 points) and REM (17.35 points), although narrow N2 budget intervals still favor the baselines (Table XXII). When the backbone, calibration, folds, and evaluation epochs are fixed, the utility-aware stopper independently improves macro-F1 by 1.78–3.92 points over the common budget range and reaches its peak with 29.1%–46.6% fewer patches than the strongest alternative. These policy-side gains are concentrated in N1 and N3, while the average reductions in W and N2 remain within 0.45 points (Table XXIV). Independently, under full observation, the PAUSE backbone achieves the highest mean ACC on all four datasets and the highest mean macro-F1 on three, while trailing AttnSleep by 0.38 points on Sleep-EDF-78. These results support a competitive classification foundation for adaptive acquisition. The three controlled decompositions are therefore complementary: the backbone provides discriminative evidence, whereas the stopper converts that evidence into a more favorable class-balanced acquisition policy. The overall

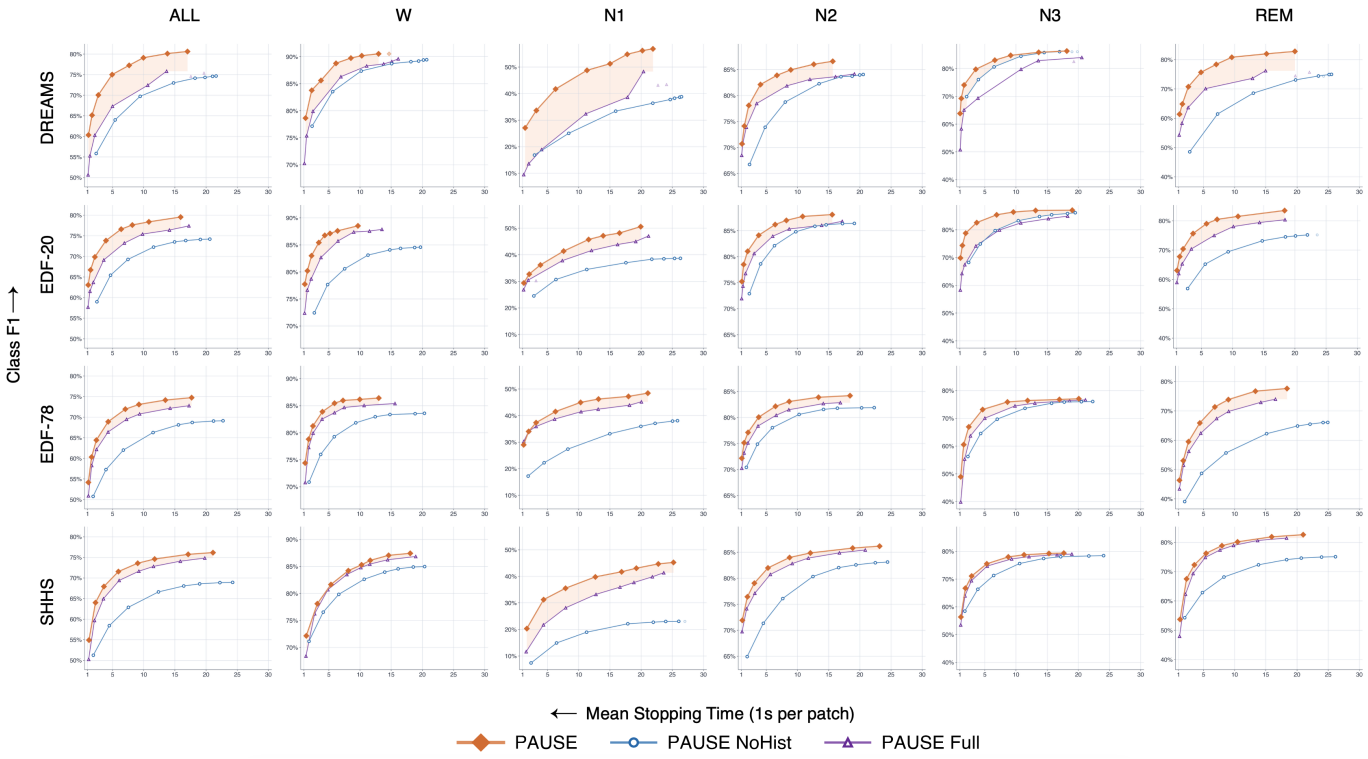



Fig. 5. Class-wise speed–accuracy frontiers of the two ablations. Rows correspond to datasets and columns to the overall macro-F1 (ALL) and the five sleep stages. The horizontal axis is the reference-stage-conditional mean stopping time $\mathbb{E}[\tau \mid c]$ of each variant, and the vertical axis is pooled class-wise F1. PAUSE NoHist removes the endpoint keys and values, and PAUSE Full trains on the full-history view alone; each variant performs its own closed-loop rollout. Solid lines join within-variant Pareto points, faint markers denote dominated operating points, and orange shading marks budgets at which PAUSE exceeds the joint upper envelope of the two ablations.

TABLE XXVII

DETAILED SLEEP-STAGING PERFORMANCE ON DREAMS-SUB, SLEEP-EDF-20, SLEEP-EDF-78, AND SHHS. WITHIN EACH DATASET AND METRIC, BOLD MARKS THE HIGHEST FOLD MEAN AND UNDERLINE THE SECOND HIGHEST. ALL ENTRIES ARE EQUAL-WEIGHT MEANS OVER 20 HELD-OUT FOLDS (%); STANDARD DEVIATIONS OF THE OVERALL METRICS ARE GIVEN IN THE MAIN PAPER.

| **Dataset** | **Method** | **Class-wise F1 (%)** | | | | | **Overall metrics** | | |
|---|---|---|---|---|---|---|---|---|---|
| | | **W** | **N1** | **N2** | **N3** | **REM** | **ACC (%)** | **MF1 (%)** | **MGm (%)** |
| DREAMS-SUB | AttnSleep (2021) | 86.12 | 38.14 | 85.61 | 85.77 | 76.75 | 82.39 | 74.48 | 82.86 |
| | LGSleepNet (2023) | 87.84 | 36.78 | 85.77 | 83.70 | 77.47 | 82.47 | 74.31 | 82.63 |
| | MSA-CNN (2026) | 85.69 | 21.90 | 84.58 | 84.21 | 71.74 | 81.50 | 69.62 | 76.93 |
| | ULW-SleepNet (2026) | **89.64** | 19.95 | 86.06 | **85.95** | 76.04 | 82.87 | 71.53 | 79.05 |
| | **PAUSE (ours)** | 88.71 | **50.80** | **86.42** | 83.01 | **81.98** | **84.79** | **78.18** | **84.64** |
| Sleep-EDF-20 | AttnSleep (2021) | 87.97 | 43.58 | 88.68 | 88.84 | 80.50 | 84.21 | 77.92 | **86.20** |
| | LGSleepNet (2023) | 87.59 | 41.88 | 88.43 | **89.25** | 79.67 | 83.91 | 77.36 | 85.59 |
| | MSA-CNN (2026) | 82.50 | 9.64 | 88.12 | 87.77 | 76.54 | 82.48 | 68.91 | 75.37 |
| | ULW-SleepNet (2026) | 87.80 | 28.48 | **88.80** | 88.78 | 78.36 | 84.12 | 74.45 | 81.74 |
| | **PAUSE (ours)** | **90.27** | **46.84** | 88.29 | 85.01 | **83.81** | **85.58** | **78.84** | 85.49 |
| Sleep-EDF-78 | AttnSleep (2021) | 87.02 | 43.00 | 84.67 | **80.01** | 74.81 | 78.47 | **73.90** | 82.77 |
| | LGSleepNet (2023) | 86.40 | 43.27 | 84.54 | 79.46 | 74.24 | 78.03 | 73.58 | **82.82** |
| | MSA-CNN (2026) | 86.95 | 35.90 | **85.06** | 79.68 | 71.53 | 78.23 | 71.82 | 80.63 |
| | ULW-SleepNet (2026) | 85.86 | 34.45 | 84.26 | 78.23 | 67.55 | 76.81 | 70.07 | 79.22 |
| | **PAUSE (ours)** | **87.79** | **45.74** | 84.61 | 71.10 | **78.35** | **79.73** | 73.52 | 81.62 |
| SHHS | AttnSleep (2021) | 85.63 | 33.07 | 86.90 | 79.81 | 79.26 | 81.73 | 72.94 | 82.87 |
| | LGSleepNet (2023) | 86.70 | 30.90 | **87.14** | 81.65 | 79.62 | 82.53 | 73.20 | 82.60 |
| | MSA-CNN (2026) | 86.44 | 0.00 | 86.89 | **82.60** | 77.96 | 82.72 | 66.78 | 71.50 |
| | ULW-SleepNet (2026) | 82.87 | 0.97 | 82.80 | 78.68 | 70.75 | 78.10 | 63.21 | 69.33 |
| | **PAUSE (ours)** | **88.31** | **42.74** | 86.63 | 78.69 | **83.04** | **83.61** | **75.88** | **83.30** |

frontier improvement reflects their combination, which reduces EEG acquisition without shifting errors toward the minority stage. At the same time, the full-observation N3 deficits and the small stopper-side W and N2 trade-offs identify concrete directions for further improvement.

## VII. Extension Experiments

### A. Online Staging

Sleep staging is a key enabling component for many downstream sleep applications because it provides an estimate of sleep state on which subsequent analysis or intervention depends. This role is especially important in real-time applications, including sleep-onset detection and stage-gated closed-loop stimulation [29]–[31]. Therefore, when designing resource-efficient sleep staging systems, it is important to preserve their ability to support different downstream tasks. An offline formulation may have higher performance but makes stage labels available only retrospectively, and thus cannot provide the during-sleep feedback required by real-time control. A causal online formulation can instead expose stage estimates during acquisition and make them available for downstream control.

MASS is a resource-efficient sleep staging framework which fixes a partial-observation scheme independently of the accumulated evidence [5]. Although it achieves resource-efficient sleep staging with a smaller amount of signal, it relies on complete context information of both preceding and following epochs. This makes it a strict offline computation model that cannot support real-time closed-loop control tasks. The proposed PAUSE is completely different from it. Its endpoint $\tau_n$ must be a stopping time with respect to $\mathcal{F}_{n,k} = \sigma(\mathcal{O}_{n,k})$. Therefore, PAUSE accepts a stricter causal information requirement in exchange for a broader operational capability: its output can become available during acquisition and can serve as an online feedback signal.

Based on this, we report MASS as the closest offline predecessor and a strong reference in this section, rather than as a member of the causal stopping frontier in the main ECTS comparison. The numerical difference below is an observed system-level gap between these two acquisition contracts, not a controlled estimate of a single causal effect.

*1) Budget translation.:* MASS controls its budget by two mask ratios rather than by a stopping time. An epoch-level ratio $r_e$ removes epochs from a window of length $L$, and a patch-level ratio $r_a$ removes patches inside each retained epoch. Because the implementation masks integer numbers of tokens, it removes $M_e = \lceil Lr_e \rceil$ epochs and $M_a = \lceil Pr_a \rceil$ patches per retained epoch. The effective mean acquisition count is therefore

$$B_{\mathrm{MASS}} = \frac{(L - M_e)(P - M_a)}{L}.$$

Table XXVIII distinguishes the nominal signal integrity reported by MASS from the effective integer-valued acquisition count used for comparison. The two systems are comparable in the amount of acquired signal, but they are completely different in terms of deployment and observation logic: PAUSE acquires a contiguous prefix and issues a decision for every epoch from at least one patch, whereas MASS will first randomly decide the signals to be collected, and offline classification can only be performed after all collection is completed.

TABLE XXVIII
Translation of the three partial-observation MASS settings retained for comparison onto the acquisition axis used in this paper. The final column accounts for integer token masking and gives the effective mean number of one-second patches acquired per epoch.

| **Nominal integrity** | $r_a$ | $r_e$ | $(1 - r_a)(1 - r_e)$ | $B_{\mathrm{MASS}}$ |
|---|---|---|---|---|
| 72% | 0.2 | 0.1 | 0.72 | 21.0 |
| 40% | 0.5 | 0.2 | 0.40 | 11.72 |
| 10% | 0.8 | 0.5 | 0.10 | 3.0 |

*2) Joint accuracy–latency comparison.:* Table XXIX reads both systems at the three effective acquisition ceilings of Table XXVIII. MASS is evaluated at the corresponding published masking setting. Within the recorded range, the PAUSE score is obtained by piecewise-linear interpolation between adjacent operating points. If a ceiling exceeds the largest recorded PAUSE budget, we instead report the most expensive observed PAUSE point that remains below the ceiling. These largest recorded budgets are 17.01, 15.92, 17.68, and 21.10 patches on the four datasets, respectively.

The same table also reports when each label becomes available. Since the bidirectional encoder of MASS consumes the complete $L$-epoch window, no label in that window can be emitted before its final epoch has been acquired. Measured from the onset of epoch $n$, its decision latency and window mean are

$$\ell_n^{\mathrm{MASS}} = P\,(L - n + 1), \qquad \bar{\ell}^{\mathrm{MASS}} = \frac{P(L+1)}{2} = 495 \text{ s}.$$

The latency ranges from 30 s for the last epoch to 960 s for the first. PAUSE instead emits the label of epoch $n$ after its acquired prefix, so its mean label-availability delay at an operating point is $\bar{\ell}^{\mathrm{PAUSE}} = \mathbb{E}[\tau]$.

The comparison is most informative as retained accuracy under added online capability. Across the eight dataset–ceiling comparisons at $B \geq 11.72$, PAUSE remains within 0.70 macro-F1 percentage points of MASS in every case; the mean difference is 0.35 points, and PAUSE is ahead by 0.30 points in one setting. At the same time, PAUSE makes each decision from causally available evidence and reduces the corresponding window-mean label-availability delay from 495 s to 11.72–21.00 s at these moderate-to-high ceilings. PAUSE therefore retains nearly all of the strong offline reference's accuracy while converting the output into an online, potentially actionable staging signal. Across the retained settings, PAUSE reduces the window-mean label-availability delay by factors of 23.6–165.0.

The difference increases to 4.13–5.77 points at the extreme three-patch ceiling. This setting corresponds to 10% nominal

TABLE XXIX

JOINT ACCURACY–LATENCY COMPARISON UNDER COMMON ACQUISITION CEILINGS. SYSTEM ROWS REPORT MACRO-F1 (%) / MEAN LABEL-AVAILABILITY DELAY (S). $\Delta\mathrm{MF1} = \mathrm{PAUSE} - \mathrm{MASS}$, SO A POSITIVE ENTRY MARKS A CEILING AT WHICH PAUSE IS AHEAD. THE DELAY RATIO IS $\bar{\ell}^{\mathrm{MASS}}/\bar{\ell}^{\mathrm{PAUSE}}$ (M/P). PAUSE USES INTERPOLATION ONLY WITHIN ITS RECORDED FRONTIER AND OTHERWISE THE MOST EXPENSIVE OBSERVED POINT BELOW THE CEILING.

| **Dataset** | **System / comparison** | **Acquisition ceiling $B$** | | |
|---|---|---|---|---|
| | | 3.0 | 11.72 | 21.0 |
| DREAMS-SUB | MASS: MF1 / delay | 75.58/495 | 80.02/495 | 80.71/495 |
| | PAUSE: MF1 / delay | 70.68/3.00 | 79.58/11.72 | 80.63/17.01 |
| | ΔMF1 | −4.90 | −0.44 | −0.08 |
| | Delay ratio (M/P) | 165.0× | 42.2× | 29.1× |
| Sleep-EDF-20 | MASS: MF1 / delay | 76.62/495 | 79.21/495 | 79.26/495 |
| | PAUSE: MF1 / delay | 71.73/3.00 | 78.60/11.72 | 79.56/15.92 |
| | ΔMF1 | −4.89 | −0.61 | **+0.30** |
| | Delay ratio (M/P) | 165.0× | 42.2× | 31.1× |
| Sleep-EDF-78 | MASS: MF1 / delay | 71.61/495 | 74.42/495 | 74.95/495 |
| | PAUSE: MF1 / delay | 65.84/3.00 | 73.72/11.72 | 74.74/17.68 |
| | ΔMF1 | −5.77 | −0.70 | −0.21 |
| | Delay ratio (M/P) | 165.0× | 42.2× | 28.0× |
| SHHS | MASS: MF1 / delay | 70.25/495 | 75.18/495 | 76.58/495 |
| | PAUSE: MF1 / delay | 66.12/3.00 | 74.59/11.72 | 76.15/21.00 |
| | ΔMF1 | −4.13 | −0.59 | −0.43 |
| | Delay ratio (M/P) | 165.0× | 42.2× | 23.6× |

signal integrity and maximizes the difference in observation method: MASS masks half of the epochs and can recover their labels from surrounding, including future, context, whereas PAUSE still stages every epoch causally from its own acquired prefix and strictly preceding evidence.

Taken together, these results demonstrate that PAUSE retains most of the accuracy of a strong offline reference while producing labels during acquisition rather than after the window completes. This distinction is consequential: causality, label availability, and evidence-adaptive acquisition are not secondary deployment details, but determine whether a staging output can actually support time-sensitive downstream intervention. By jointly deciding what evidence to acquire, when sufficient evidence has been observed, and when a label can be released, PAUSE transforms sleep staging from retrospective inference over completed windows into an actionable closed-loop sensing process.

### *B. Analysis of the Per-Prefix Calibration*

In PAUSE, the stopper does not act on the evidence directly. It compares an immediate classification risk against an expected continuation risk, and both quantities are expectations taken under the calibrated posterior $\boldsymbol{\pi}_{n,k}^{\mathrm{cal}}$. The immediate risk is linear in that posterior, so an overconfident posterior lowers it proportionally, and the rule stops earlier than the true risk warrants. This dependence is not particular to PAUSE. Any rule that minimizes an expected cost, rather than simply ranked classes, requires probability estimates on the correct scale [32]. Modern neural classifiers are in turn known to be systematically overconfident, and temperature scaling is the standard post-hoc remedy. It can repair the probabilities while leaving the predicted labels intact [33]. This requirement is explicit in the finite-horizon framework we follow, which restricts its risk-based comparison to models whose sufficient statistic is well calibrated because overconfidence inflates the estimated risk beyond its calibrated level [14], [15]. We index the temperature by the prefix because the continuation-risk model is itself fitted per step, there as in PAUSE, so the probability scale has to be correct at each $k$ rather than on average. Therefore, calibration is a prerequisite of the stopping rule rather than a cosmetic refinement of the classifier. We examine the fitted temperatures themselves, obtained under the cross-fitted out-of-fold procedure described in the cross-validation protocol above. Every number below is computed from the three training variants (PAUSE, w/o Paired-History, and w/o Endpoint Memory) on all four datasets.

*1) Temperature dependence on the prefix length.:* Since each fold contributes one temperature per prefix, the prefix dependence can be summarized by the within-fold contrast $\Delta = T_1 - T_{30}$. The 20 fold-specific temperatures are leave-one-fold-out estimates whose source sets overlap heavily, since any two of them share 18 of their 19 source folds. Their naive dispersion is consequently a deflated measure of sampling variability. We therefore report the jackknife standard error $\mathrm{SE}_{\mathrm{jack}} = \frac{n-1}{\sqrt{n}}\, s$ with $n = 20$, where $s$ is the naive standard deviation of $\Delta$ across folds, which is $4.25$ times the naive value. We regard $|\Delta|/\mathrm{SE}_{\mathrm{jack}} \geq 3$ as evidence that the prefix dependence is not an artifact of fitting noise.

Table XXX shows that the relationship between temperature and prefix length is dataset dependent, and that its direction is not shared across datasets. On Sleep-EDF-78 the fitted temperatures at the shortest and longest prefixes are similar, but they remain above one at every prefix, and the held-out results below show improved NLL at all 30 prefixes together with a reduction in ECE after calibration.

These results show that no single monotonic relationship be-

TABLE XXX

PREFIX DEPENDENCE OF THE FITTED TEMPERATURE. $T_1$ AND $T_{30}$ ARE FOLD-AVERAGED TEMPERATURES AT THE SHORTEST AND LONGEST PREFIX; $\Delta = T_1 - T_{30}$ IS COMPUTED WITHIN EACH FOLD AND THEN AVERAGED. $\mathrm{SE}_{\mathrm{jack}}$ IS THE JACKKNIFE STANDARD ERROR OF $\Delta$, WHICH CORRECTS FOR THE OVERLAP BETWEEN LEAVE-ONE-FOLD-OUT ESTIMATES. BOLD MARKS $|\Delta|/\mathrm{SE}_{\mathrm{jack}} \geq 3$.

| **Variant** | **Dataset** | $T_1$ | $T_{30}$ | $\Delta$ | $\mathrm{SE}_{\mathrm{jack}}$ | $\lvert\Delta\rvert/\mathrm{SE}$ |
|---|---|---|---|---|---|---|
| PAUSE | DREAMS-SUB | 1.391 | 1.152 | +0.239 | 0.052 | **4.6** |
| | Sleep-EDF-20 | 1.395 | 1.248 | +0.147 | 0.037 | **4.0** |
| | Sleep-EDF-78 | 1.290 | 1.281 | +0.010 | 0.018 | 0.5 |
| | SHHS | 1.261 | 1.387 | −0.126 | 0.024 | **5.3** |
| w/o Paired-History | DREAMS-SUB | 1.523 | 1.301 | +0.223 | 0.028 | **7.9** |
| | Sleep-EDF-20 | 1.533 | 1.352 | +0.181 | 0.023 | **7.9** |
| | Sleep-EDF-78 | 1.470 | 1.478 | −0.008 | 0.045 | 0.2 |
| | SHHS | 1.317 | 1.406 | −0.089 | 0.028 | **3.2** |
| w/o Endpoint Memory | DREAMS-SUB | 1.298 | 1.054 | +0.244 | 0.132 | 1.8 |
| | Sleep-EDF-20 | 1.354 | 1.245 | +0.109 | 0.106 | 1.0 |
| | Sleep-EDF-78 | 1.097 | 1.165 | −0.068 | 0.041 | 1.6 |
| | SHHS | 1.085 | 1.195 | −0.109 | 0.024 | **4.5** |

tween temperature and prefix length transfers across datasets: the endpoint contrast is significantly positive on DREAMS-SUB and Sleep-EDF-20, significantly negative on SHHS, and indistinguishable from zero on Sleep-EDF-78. A predefined schedule would therefore be misspecified in sign on at least one dataset, and on the three datasets with a significant contrast a single shared temperature cannot be correct at both endpoints. We estimate one temperature per prefix from prefix-specific held-out predictions and apply this protocol uniformly across datasets and variants.

*2) Sensitivity of the fitted temperature to its source set.:* A practical question about any cross-fitted calibration is how strongly the fitted temperature depends on the composition of its source set. The 20 leave-one-fold-out temperature vectors differ precisely by which fold is excluded, so their spread at each prefix measures this directly. For PAUSE, the range across the 20 vectors divided by their mean, taken at the median prefix, is $8.3\%$ on DREAMS-SUB, $3.7\%$ on Sleep-EDF-20, $2.7\%$ on Sleep-EDF-78 and $3.9\%$ on SHHS; the largest single-prefix value on any dataset is $9.3\%$, an absolute range of $0.107$ in $T$. Deleting an entire source fold therefore moves the fitted temperature by less than a tenth of its value, so the calibration is stable with respect to its fitting sample.

*3) The fitted temperatures transfer across subjects.:* The temperature is obtained by minimizing negative log-likelihood, so NLL is the criterion that corresponds to the fitted objective; while expected calibration error is a coarser binned statistic.

Table XXXI shows that for PAUSE, the held-out NLL improves at all 30 prefix lengths on all four datasets, and the ECE falls from $4.28$–$5.37\%$ to $2.14$–$3.82\%$. The transfer is thus consistent at the level of the fold population. It is not universal at the level of the individual subject: between one and seven of the 20 held-out folds per dataset show a worse prefix-averaged NLL, most frequently on DREAMS-SUB, which is also the dataset with the fewest subjects and the smallest ECE reduction. We therefore state the property as a population-level one and do not claim that a cross-fitted temperature improves every recording. A second observation concerns the variants. The model trained on full histories alone is the least well calibrated before scaling on all four datasets, with an ECE of $5.28$–$7.47\%$ against $4.28$–$5.37\%$ for PAUSE, and it requires uniformly larger temperatures. Paired-history training therefore appears to yield not only a better closed-loop frontier but also a less overconfident posterior. The endpoint-memory ablation exhibits a less stable transfer pattern. Although temperature scaling reduces the average NLL at nearly all prefix lengths on three datasets, the improvements are consistently smaller than those of PAUSE. The instability is most evident on DREAMS-SUB, where only 16 of the 30 prefixes improve and 11 of the 20 held-out folds show worse prefix-averaged NLL after scaling. These results suggest that endpoint memory helps produce posterior distributions whose calibration transfers more reliably across held-out folds.

*4) Calibration-Induced Decision Changed.:* Per-prefix calibration is necessary for the risk-based stopper because the finite-horizon Bellman recursion evaluates stopping and continuation risks under the predicted posterior, and an overconfident posterior can underestimate the immediate stopping risk. The comparison below uses the three decision rules of the Prefix-Wise Curves subsection: the head emits prior-free evidence $s_{n,k}$, the calibrated MAP rule (3) reinstates the empirical prior after temperature scaling, and the pre-calibration reference is the $T_k = 1$ rule (4). In that prior-aware MAP rule, temperature scaling acts on the evidence score while leaving the empirical class prior unchanged. Comparing the minority stage N1 with N2, the calibrated MAP rule selects N1 when

$$\begin{aligned} &\frac{s(\mathrm{N1}) - s(\mathrm{N2})}{T_k} > \log \frac{\widehat{\pi}_{\mathrm{N2}}}{\widehat{\pi}_{\mathrm{N1}}} \\ \iff \quad &s(\mathrm{N1}) - s(\mathrm{N2}) > T_k \log \frac{\widehat{\pi}_{\mathrm{N2}}}{\widehat{\pi}_{\mathrm{N1}}}. \end{aligned} \tag{5}$$

TABLE XXXI

CALIBRATION QUALITY ON HELD-OUT FOLDS, AVERAGED OVER THE 30 PREFIXES AND THE 20 FOLDS. "$k$ WITH NLL ↓" COUNTS PREFIX LENGTHS AT WHICH THE FOLD-AVERAGED HELD-OUT NLL IMPROVES; "FOLDS NLL ↑" COUNTS HELD-OUT FOLDS WHOSE PREFIX-AVERAGED NLL BECOMES WORSE.

| **Variant** | **Dataset** | ECE (%) | | NLL | | $k$ with NLL ↓ | folds NLL ↑ |
|---|---|---|---|---|---|---|---|
| | | before | after | before | after | | |
| PAUSE | DREAMS-SUB | 4.28 | 3.82 | 0.5095 | 0.5007 | 30/30 | 7/20 |
| | Sleep-EDF-20 | 4.70 | 2.86 | 0.4868 | 0.4700 | 30/30 | 4/20 |
| | Sleep-EDF-78 | 5.37 | 2.85 | 0.6185 | 0.6001 | 30/30 | 6/20 |
| | SHHS | 5.15 | 2.14 | 0.5668 | 0.5404 | 30/30 | 1/20 |
| w/o Paired-History | DREAMS-SUB | 6.21 | 5.57 | 0.5479 | 0.5225 | 30/30 | 9/20 |
| | Sleep-EDF-20 | 6.28 | 4.55 | 0.5312 | 0.4979 | 30/30 | 6/20 |
| | Sleep-EDF-78 | 7.47 | 4.06 | 0.6612 | 0.6151 | 30/30 | 4/20 |
| | SHHS | 5.28 | 2.68 | 0.5496 | 0.5186 | 30/30 | 5/20 |
| w/o Endpoint Memory | DREAMS-SUB | 5.41 | 5.37 | 0.6488 | 0.6481 | 16/30 | 11/20 |
| | Sleep-EDF-20 | 5.57 | 4.46 | 0.6612 | 0.6491 | 30/30 | 6/20 |
| | Sleep-EDF-78 | 4.09 | 3.32 | 0.7741 | 0.7692 | 29/30 | 6/20 |
| | SHHS | 3.64 | 2.10 | 0.7152 | 0.7074 | 30/30 | 4/20 |

TABLE XXXII

DECISION-LAYER EFFECT OF PER-PREFIX CALIBRATION ON THE DIAGNOSTIC MAP RULE (3), AVERAGED OVER THE 30 PREFIXES AND 20 FOLDS; THE PAUSE STOPPER INSTEAD OUTPUTS THE CLASS-WEIGHTED BAYES ACTION DEFINED IN THE MAIN MANUSCRIPT. $\widehat{\boldsymbol{\pi}}$ IS THE FOLD-AVERAGED TRAINING-SPLIT PRIOR. THE ADDED MARGIN IS $(\overline{T}-1)\log(\widehat{\pi}_{\mathrm{N2}}/\widehat{\pi}_{\mathrm{N1}})$, AND RECALL AND MACRO-F1 CHANGES ARE REPORTED IN PERCENTAGE POINTS AFTER CALIBRATION.

| **Dataset** | $\widehat{\pi}_{\mathrm{N1}}$ (%) | $\widehat{\pi}_{\mathrm{N2}}/\widehat{\pi}_{\mathrm{N1}}$ | $\overline{T}$ | Added margin | $\Delta R_{\mathrm{N1}}$ | $\Delta R_{\mathrm{N2}}$ | ΔMF1 |
|---|---|---|---|---|---|---|---|
| DREAMS-SUB | 7.4 | 5.59 | 1.215 | 0.371 | −5.26 | +1.56 | −0.88 |
| Sleep-EDF-20 | 7.0 | 6.37 | 1.285 | 0.527 | −6.42 | +1.79 | −1.08 |
| Sleep-EDF-78 | 12.7 | 3.33 | 1.293 | 0.352 | −5.74 | +2.67 | −1.58 |
| SHHS | 3.2 | 13.88 | 1.353 | 0.927 | −13.90 | +2.68 | −2.65 |

Because the fitted temperatures satisfy $T_k > 1$, calibration increases the evidence margin required for N1 by $(T_k - 1)\log(\widehat{\pi}_{\mathrm{N2}}/\widehat{\pi}_{\mathrm{N1}})$, with a larger shift under stronger class imbalance or a higher temperature. This effect is specific to the prior-aware decision layer: the maximum-likelihood prediction $\arg\max_c s(c)/T_k$ is invariant to any positive temperature because it contains no prior term.

Table XXXII confirms the predicted decision shift. Under this diagnostic rule, calibration reduces N1 recall by 5.26 to 13.90 percentage points, increases N2 recall by 1.56 to 2.68 points, and lowers MAP macro-F1 by 0.88 to 2.65 points. The effect is most pronounced on SHHS, which combines the largest N2-to-N1 prior ratio with the highest mean temperature. This loss does not indicate poorer probability estimation: the N1-specific calibration error decreases from 2.88% to 2.58%, 2.90% to 2.63%, 3.63% to 2.98%, and 1.45% to 1.16% on DREAMS-SUB, Sleep-EDF-20, Sleep-EDF-78, and SHHS, respectively. The calibrated posterior is therefore a more reliable probabilistic description of N1, while the unchanged imbalanced prior makes an unweighted MAP rule less sensitive to that minority stage.

### C. Illustrative Stop-versus-Continue Risk Trajectories

Figure 6 visualizes the two risks evaluated by the PAUSE stopper along five randomly selected DREAMS-SUB epochs, one from each ground-truth sleep stage. The orange solid curve denotes the immediate stopping risk $G^{\mathrm{st}}_{n,k}$, namely the minimum expected class-weighted misclassification cost under the calibrated posterior. The green dashed curve denotes the continuation risk $G^{\mathrm{cont}}_{n,k}$, which combines the acquisition cost $\eta = 0.02$ with the GP estimate of the next-step cost-to-go. PAUSE stops at the first prefix satisfying $G^{\mathrm{st}}_{n,k} \le G^{\mathrm{cont}}_{n,k}$, as marked by the filled circles.

This comparison follows the finite-horizon risk decomposition used by FIRMBOUND, in which backward induction determines whether the immediate Bayes risk is no greater than the expected risk of continuing acquisition [15]. PAUSE instantiates this principle with a calibrated posterior conditioned on the acquired prefix and endpoint history, and replaces the class-neutral error penalty with the class-weighted cost matrix used by its utility-aware decision rule. Moreover, the continuation risk is estimated by a time-indexed sparse GP. The resulting crossings therefore visualize a fitted finite-horizon stopping rule.

To be noticed, risk is not required to decrease monotonically

with the prefix length. A newly acquired patch can shift posterior mass toward or away from competing stages, while the continuation risk also changes with the posterior state, the remaining horizon, and the GP fitted at that prefix. Consequently, transient fluctuations and later reversals are expected. Only the first crossing affects the online policy.

### D. Edge Deployment Experiment

Since the backbone in PAUSE is causal, the acquisition policy admits exact incremental decoding: each newly acquired one-second patch is processed as a single-token forward pass that reuses the cached keys and values of the current epoch's prefix together with the endpoint memories of the preceding epochs. We verified that this streaming path reproduces the training-time forward pass to within $4.3 \times 10^{-6}$ in logit space, so the deployed computation is numerically the same model that produces the frontiers reported above. One stopping decision therefore costs $2.09$ MMACs ($4.19$ MFLOPs) over a model of $2.0$M parameters ($7.6$ MiB in single precision), and the key–value cache occupies $434$ KiB.

We exported this single-step graph to ONNX and measured it on a MediaTek Dimensity 9500 (MT6993, Android 16) with ONNX Runtime 1.28 restricted to a single CPU thread in single precision, using no neural accelerator, no GPU, and no quantization to simulate a strict edge deployment. Table XXXIII reports the per-decision latency in thermal steady state. Under back-to-back execution the median latency is $2.14$ ms on the prime core and $3.32$ ms on the efficiency cluster. We additionally report the regime that matches actual deployment, in which patches arrive at $1$ Hz: there the frequency governor never ramps up and stays near $1.6$ GHz, so the median latency rises to $9.44$ ms. Even under this most conservative setting the decision latency is $105\times$ shorter than the one-second patch interval, and an entire 30-s epoch at the operating point $\mathbb{E}[\tau] = 8$ requires $76$ ms of computation, a duty cycle of $0.25\%$. These measurements support streaming inference at the one-second acquisition cadence on the tested mobile platform. They measure computation and latency, not the energy of the acquisition node or the combined sensor–host system.

The backward-induction stopping rule adds a Gaussian-process regressor per time step that maps the current posterior to the continuation risk. Only the predictive mean enters the decision, and for a whitened variational strategy it is closed-form, $\mu(x) = c + k(x, Z)^\top \alpha$ with $\alpha = L^{-\top} m$ precomputed offline. The deployed stopper therefore stores only $Z \in \mathbb{R}^{400\times 5}$, $\alpha \in \mathbb{R}^{400}$, the constant mean and the kernel hyperparameters per prefix: $0.28$ MB in single precision over all $29$ steps, or $3.5\%$ of the backbone. The dense $400 \times 400$ Cholesky factors of the variational covariance enter only the predictive variance and are discarded at export. Across $14{,}848$ predictions spanning all $29$ steps this closed form agrees with the GPyTorch pipeline to a relative error of $8.3 \times 10^{-5}$ and produces no stop/continue disagreements. One decision costs $4.42$ kMACs, $0.21\%$ of the backbone. On the same device and in the same thermal state, the head takes $0.028$ ms on the prime core and $0.036$ ms on the efficiency cluster, against $2.14$ ms and $3.32$ ms for the backbone: the stopping rule accounts for $1.1$–$1.3\%$ of per-decision latency. The GP stopping head therefore adds little computation and latency relative to the backbone in this measurement; total energy accounting additionally requires sensing, communication, and standby/wake-up costs.

## VIII. Sensitivity to the Inference Window Length

We vary the inference window length $L \in \{16, 32, 64\}$ (8, 16, and 32 minutes of recording time) on four datasets, keeping each epoch at 30 seconds. The backbone trained at $L = 32$, normalization, prior, per-prefix calibration, and GP stopper remain frozen. Each configuration performs its own closed-loop rollout: the GP selects the stopping time, and the acquired endpoint becomes history for subsequent epochs until the window resets. All lengths score the same held-out epochs once across the 20 folds; macro-F1 is computed from pooled confusion counts.

Under the same acquisition budget of 10 seconds, linear interpolation between observed Pareto points without extrapolation, PAUSE at $L = 32$ achieves macro-F1 scores of 79.13%, 78.12%, 73.28%, and 73.94% on DREAMS-SUB, Sleep-EDF-20, Sleep-EDF-78, and SHHS, respectively. On DREAMS-SUB, the corresponding scores are 77.65% for $L = 16$ and 75.95% for $L = 64$. Across the four datasets, shortening the window to $L = 16$ reduces macro-F1 by 1.46–1.64 percentage points, and extending it to $L = 64$ reduces macro-F1 by 1.90–3.92 points, with the largest loss on SHHS. The architecture thus accommodates inference at $L = 16$, 32, and 64 with the same trained parameters, with a modest performance decrease at the same budget. The smaller degradation at $L = 16$ is consistent with shorter histories remaining within the training range, whereas $L = 64$ introduces longer memory contexts and relative epoch offsets beyond that range. At the same 10-second budget, $L = 16$ and $L = 64$ still outperform the best evaluated ECTS baseline for each dataset by 9.66–12.34 and 7.28–10.64 macro-F1 percentage points, respectively. These results support PAUSE's effectiveness across the tested inference-window configurations without retraining.

## IX. Local Reliability of Endpoint Memory near Spectral Changes

The endpoint memory at $\tau_m$ aggregates the acquired prefix of epoch $m$ and its available preceding context. It does not require stationary EEG. In this section, we additionally examine two relative cases: a change observed before stopping tests whether the memory can use that observation, whereas a change after stopping concerns information forgone by early acquisition termination. We examine both cases through their effects on the immediately following epoch.

### A. Experimental Setup

We focus this analysis on our two largest datasets, Sleep-EDF-78 and SHHS, across 20 held-out folds and all eight acquisition costs, using the $L = 32$ closed-loop trajectories with the trained model, calibration, and GP stopper fixed.

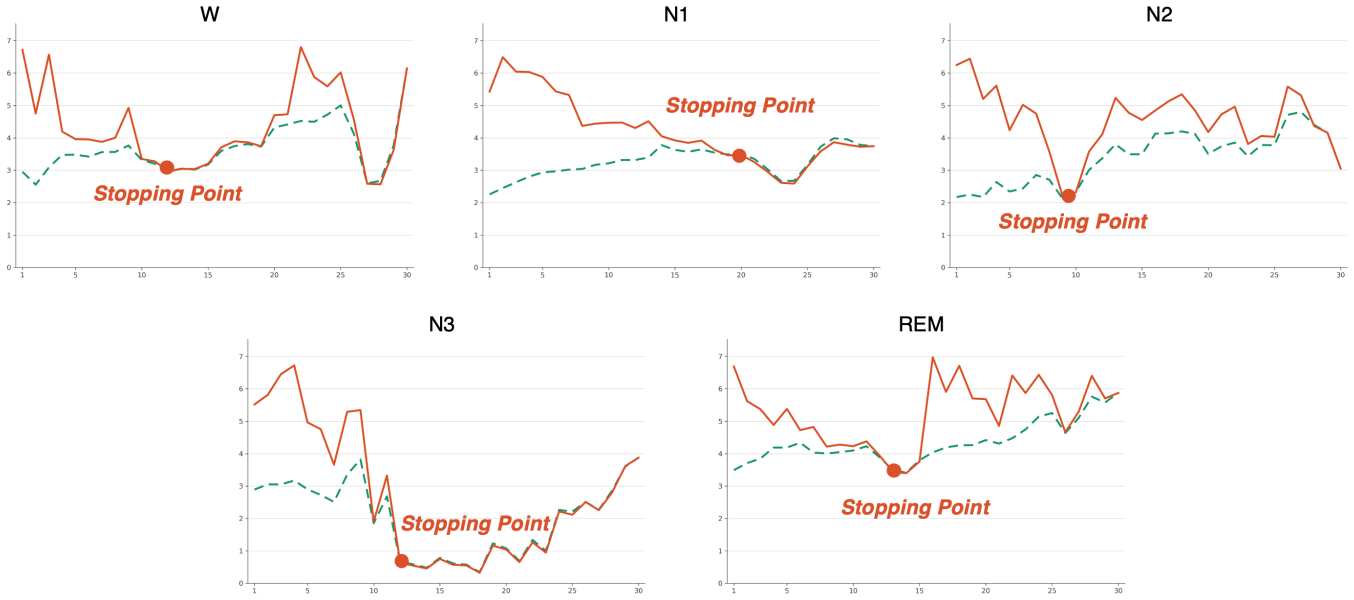



Fig. 6. Illustrative stop-versus-continue risk trajectories on DREAMS-SUB at acquisition cost $\eta = 0.02$. Panel titles indicate ground-truth sleep stages. The orange solid and green dashed curves denote the immediate stopping risk $G^{\mathrm{st}}_{n,k}$ and the fitted continuation risk $G^{\mathrm{cont}}_{n,k}$, respectively. Each filled circle marks the first prefix at which $G^{\mathrm{st}}_{n,k} \le G^{\mathrm{cont}}_{n,k}$. Prefixes after that point are shown only as counterfactual offline diagnostics and are unavailable to the deployed acquisition policy.

TABLE XXXIII
PER-DECISION INFERENCE LATENCY OF PAUSE ON A MEDIATEK DIMENSITY 9500, MEASURED WITH ONNX RUNTIME 1.28 ON A SINGLE CPU THREAD IN SINGLE PRECISION.

| Regime | Core cluster | Freq. (GHz) | Median (ms) | p90 (ms) | p99 (ms) |
|---|---|---|---|---|---|
| Back-to-back | C1-Ultra (prime) | 2.75 | 2.14 | 2.40 | 6.96 |
| | C1-Premium | 2.60 | 2.46 | 2.83 | 3.39 |
| | C1-Pro (efficiency) | 2.30 | 3.32 | 3.99 | 6.32 |
| 1 Hz duty cycle | Scheduler-assigned | 1.56 | 9.44 | 12.56 | 18.80 |
| | C1-Pro (efficiency) | 1.72 | 8.97 | 12.43 | 13.66 |

For consecutive one-second patches, we measure spectral change as

$$D_{m,j} = \frac{\|\mathbf{v}_{m,j} - \mathbf{v}_{m,j-1}\|_2}{\sqrt{128}}, \qquad j = 2, \ldots, 30, \tag{6}$$

where $\mathbf{v}_{m,j}$ is the 128-dimensional log-PSD vector normalized using training statistics. High and low change are defined by training-fold quantiles, $D \ge q_{90}$ and $D \le q_{50}$, respectively. We examine changes immediately before stopping ($D_{m,\tau_m}$) and after stopping ($D_{m,\tau_m+1}$) separately. For each $\eta$ and change location, we select at most one source epoch per window: the eligible epoch with the largest boundary-change score, provided it reaches $q_{90}$. Each source must have an immediately following epoch in the same recording and window. We match each source, where possible, to a low-change control from the same participant, $\eta$, and acquisition length without replacement.

Keeping all other historical endpoints and the next epoch's EEG fixed, we compare the source endpoint at $\tau_m$ with its complete endpoint at 30 seconds. One-second updates additionally compare $\tau_m - 1 \to \tau_m$ before stopping and $\tau_m \to \tau_m + 1$ after stopping. We measure the resulting accuracy gains at each next-epoch prefix $k = 1, \ldots, 30$ and average them over prefixes, without rerunning the stopping policy. These offline comparisons can quantify historical information value.

### B. Change Coverage under Early Stopping

We use all eligible source epochs to compute the occurrence rates. Change coverage counts high-change transitions within the acquired prefix, while missed-change sources are epochs with at least one such transition after stopping. The first-missed-step rate is calculated among early-stopped epochs.

Fig. 8 shows that shorter acquisition captures fewer high-change transitions and leaves more epochs with changes in the unobserved suffix. As mean acquisition decreases from approximately 18–21 seconds to about one second, the proportion of epochs with an unobserved high change rises from approximately 42–51% to 88%, while the first-missed-step rate stays near 10%. Earlier stopping therefore leaves more changes in the longer unobserved suffix.

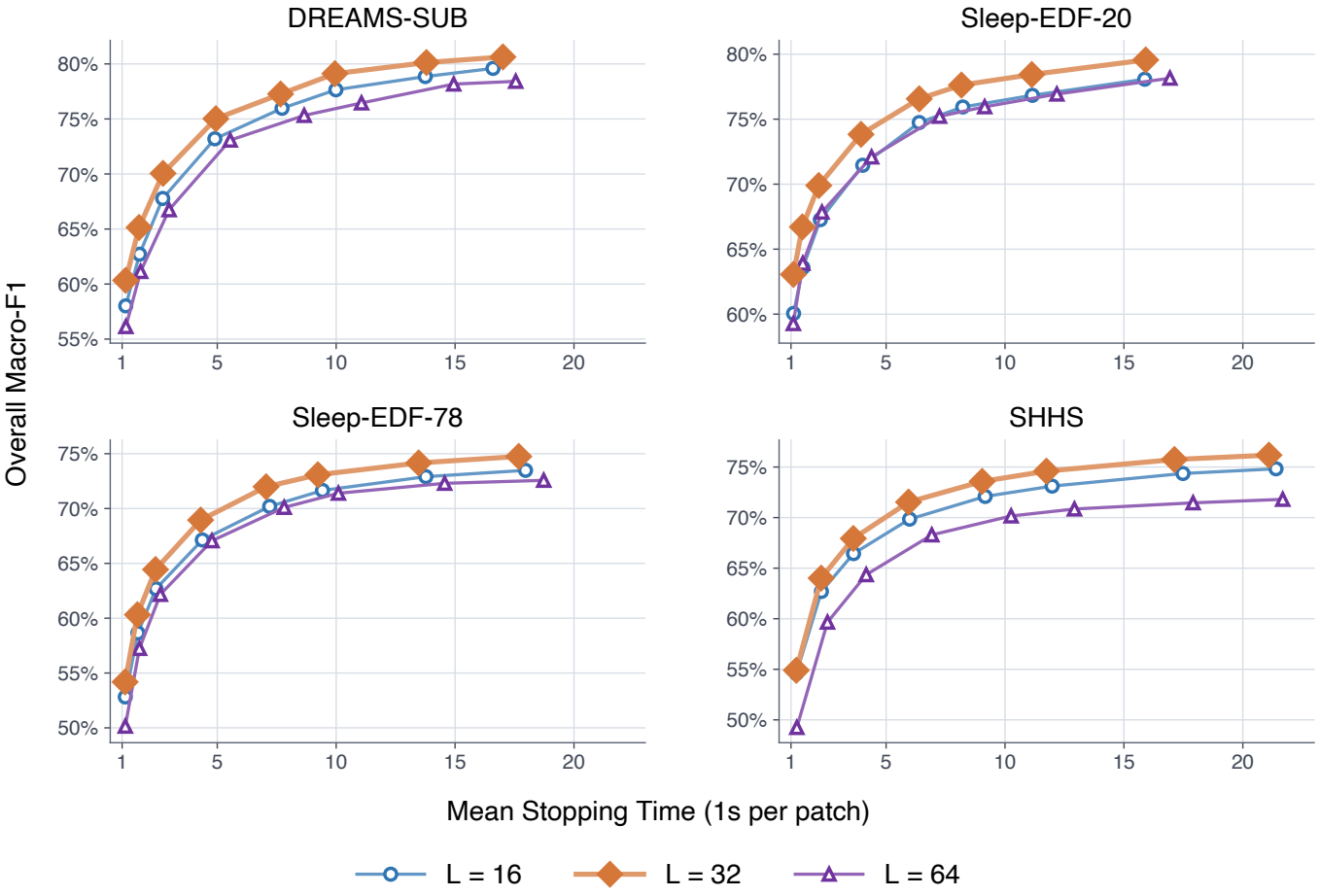



Fig. 7. Speed–accuracy frontiers of complete PAUSE on four datasets with GP stopping at three inference window lengths. Markers show the eight acquisition-cost settings per length; lines join empirical Pareto points. Upper-left is better.

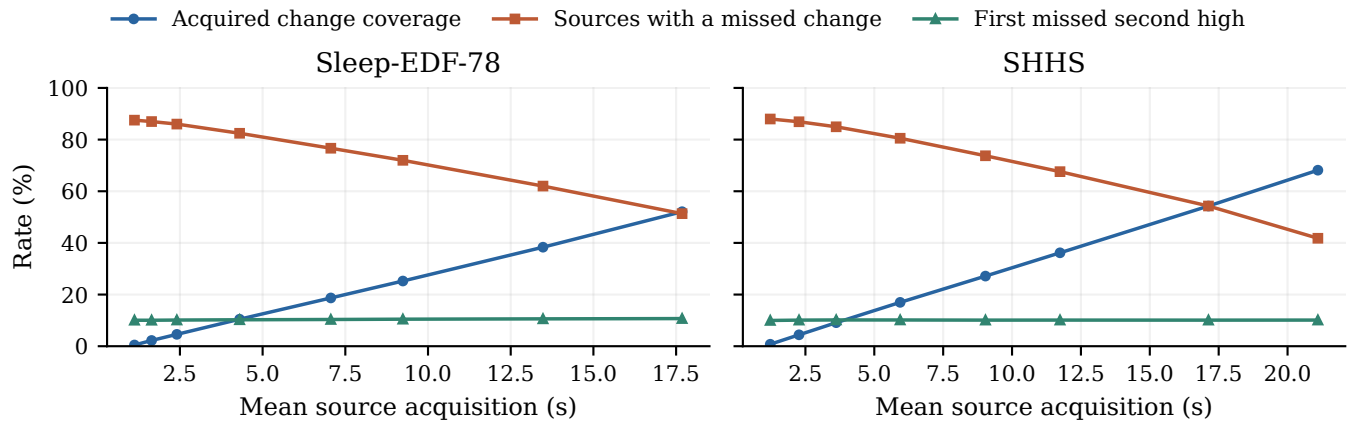



Fig. 8. Acquisition and spectral-change occurrence across all eight costs. Each marker denotes one cost. Coverage counts high-change transitions, missed-change sources count eligible epochs, and the first-missed-step rate conditions on early-stopped sources.

### C. Observed Changes Support Endpoint Memory

To test whether endpoint memory carries observed rapid changes forward, we use the selected source epochs whose last observed transition is large ($D_{m,\tau_m} \geq q_{90}$, $2 \leq \tau_m \leq 29$). For each source epoch $m$, we compare its endpoint at $\tau_m$ with an endpoint recomputed from one second less EEG, at $\tau_m - 1$, and measure accuracy in its immediate successor $m+1$. The next epoch's EEG and all other historical endpoints remain fixed. We apply the same comparison to matched low-change controls. This comparison can measure the contribution of the final observed second through endpoint memory.

We perform this comparison at all eight $\eta$ values. In the matched high-change groups, the mean next-epoch prefix accuracy gains are positive at seven of eight settings on both Sleep-EDF-78 and SHHS. At the six settings with $\eta \geq 0.02$, gains range from 0.36–1.55 and 0.31–1.15 percentage points, respectively. The two lowest-cost settings are uncertain. Table XXXIV gives the selected and matched sample counts at the representative $\eta = 0.06667$, where the matched high-change gains are 0.55 and 0.95 points. On SHHS, the gain exceeds that of matched low-change controls by 0.70 percentage points. These results support the endpoint-memory design by showing that observed changes can provide useful context for subsequent predictions.

### D. Unobserved Changes and the Acquisition Trade-off

To quantify useful history forgone by early stopping, we separately select source epochs with a large change at the first unobserved second ($D_{m,\tau_m+1} \geq q_{90}$, $\tau_m < 30$). For each selected source $m$ and each matched low-change control, we extend only that epoch from $\tau_m$ to 30 seconds and evaluate its immediate successor $m+1$, keeping the successor's EEG and all other historical endpoints fixed. These sources are selected separately from the before-stopping group. Table XXXIV reports their sample counts at $\eta = 0.06667$.

Across all eight $\eta$ values on both datasets, mean completion gains have positive point estimates in both matched high-change groups and their low-change controls. The matched high–low difference is positive at every setting. At the representative $\eta = 0.06667$, completion gains are 1.94 and 1.40 points on Sleep-EDF-78 and SHHS, with high–low differences of 1.15 and 0.61 points (Table XXXIV). Earlier stopping therefore saves acquisition while sometimes omitting context useful for subsequent predictions. These effects quantify the value of unacquired information, rather than damage to the encoding of observed EEG.

The after-stopping curves in Fig. 9 show that, at every tested $\eta$ on each dataset, the complete-history accuracy gain at the next epoch's 30-second prefix is smaller than at its one-second prefix. For example, at $\eta = 0.06667$, the gains decrease from 3.05 to 1.40 points on Sleep-EDF-78 and from 2.25 to 0.39 on SHHS. These curves use all selected high-change sources, while Table XXXIV uses matched subsets. Thus, new current-epoch evidence reduces the influence of missing history, although a residual gap can remain. The closed-loop frontier comparison incorporates this trade-off: PAUSE still outperforms the evaluated ECTS baselines on Sleep-EDF-78 and SHHS across comparable acquisition budgets.

## X. Direct Evaluation of Policy-Induced History Shift

### A. Controlled History-Switch Protocol

We directly measure how replacing complete histories with stopper-produced histories affects the backbone, and whether paired-history training reduces this sensitivity. We evaluate the existing Full-only and Paired backbones, both retaining endpoint memory, on all four datasets, 20 held-out folds, and eight acquisition costs. The window length is $L = 32$. Checkpoints, normalization, calibration, and GP stoppers remain frozen throughout this diagnostic.

For each cost, histories come from the PAUSE reference system with its frozen calibration and utility-aware GP policy. Both backbones receive exactly the same endpoint positions and recompute their own causal historical representations. We compare four conditions: Full-only with complete or policy histories $(A, B)$, and Paired with complete or the same policy histories $(C, D)$. Complete histories retain all 30 patches of each preceding epoch. At each current prefix $k$, all four conditions receive the same current EEG prefix. All diagnostic predictions use uncalibrated ML, $\arg\max_c s_{n,k}(c)$, so temperature scaling, prior injection, and class-weighted terminal actions do not affect the classification comparison.

We score each original epoch once, excluding the first position of each window because it has no preceding history. Within each fold, we compute macro-F1 at each of the 30 prefixes and then average over prefixes. Effects are paired within folds before reporting their mean and population standard deviation. Based on the $A, B, C, D$ defined before, the policy-history advantage and history-switch interaction are

$$\begin{aligned} G_{\text{policy}} &= D - B, \qquad G_{\text{full}} = C - A, \\ I &= (A - B) - (C - D) \\ &= G_{\text{policy}} - G_{\text{full}}. \end{aligned} \tag{7}$$

Positive $I$ indicates a smaller history-switch loss for Paired, after accounting for its advantage under complete histories.

### B. Predictive Effects of Real Policy Histories

Across all 32 dataset–cost combinations, replacing complete with policy histories reduces Full-only prefix-averaged macro-F1 by 0.89–12.02 percentage points. Paired training achieves higher mean macro-F1 on the same policy histories by 0.42–2.91 points and reduces the history-switch loss by 0.27–2.10 points at every operating point (Fig. 10). Its complete-history advantage is 0.15–0.96 points. This additional gain under policy histories supports reduced sensitivity to history truncation.

Uniform endpoints in the random training view have an expected length of 15.5 seconds, whereas real preceding endpoints average only 1.11–1.23 seconds at $\eta = 0.5$. Gains despite this mismatch support the use of paired-history training to improve robustness to policy-generated truncation. On SHHS,

TABLE XXXIV

REPRESENTATIVE EFFECTS AT $\eta = 0.06667$, AVERAGED OVER THE NEXT EPOCH'S 30 PREFIXES. COUNTS GIVE SELECTED HIGH-CHANGE SOURCE EPOCHS (MATCHED SUBSET SIZES IN PARENTHESES), WITH ONE LOW-CHANGE CONTROL PER MATCHED SOURCE. EFFECTS USE MATCHED GROUPS AND ARE IN ACCURACY PERCENTAGE POINTS.

| Effect | Dataset | Selected high (matched) | High change | Low change | High minus low |
|---|---|---|---|---|---|
| Observed last-second update (before) | Sleep-EDF-78 | 3,893 (3,870) | 0.55 | 0.12 | 0.42 |
| | SHHS | 8,720 (8,598) | 0.95 | 0.25 | 0.70 |
| Complete-history effect (after) | Sleep-EDF-78 | 4,785 (4,771) | 1.94 | 0.78 | 1.15 |
| | SHHS | 9,565 (9,480) | 1.40 | 0.78 | 0.61 |

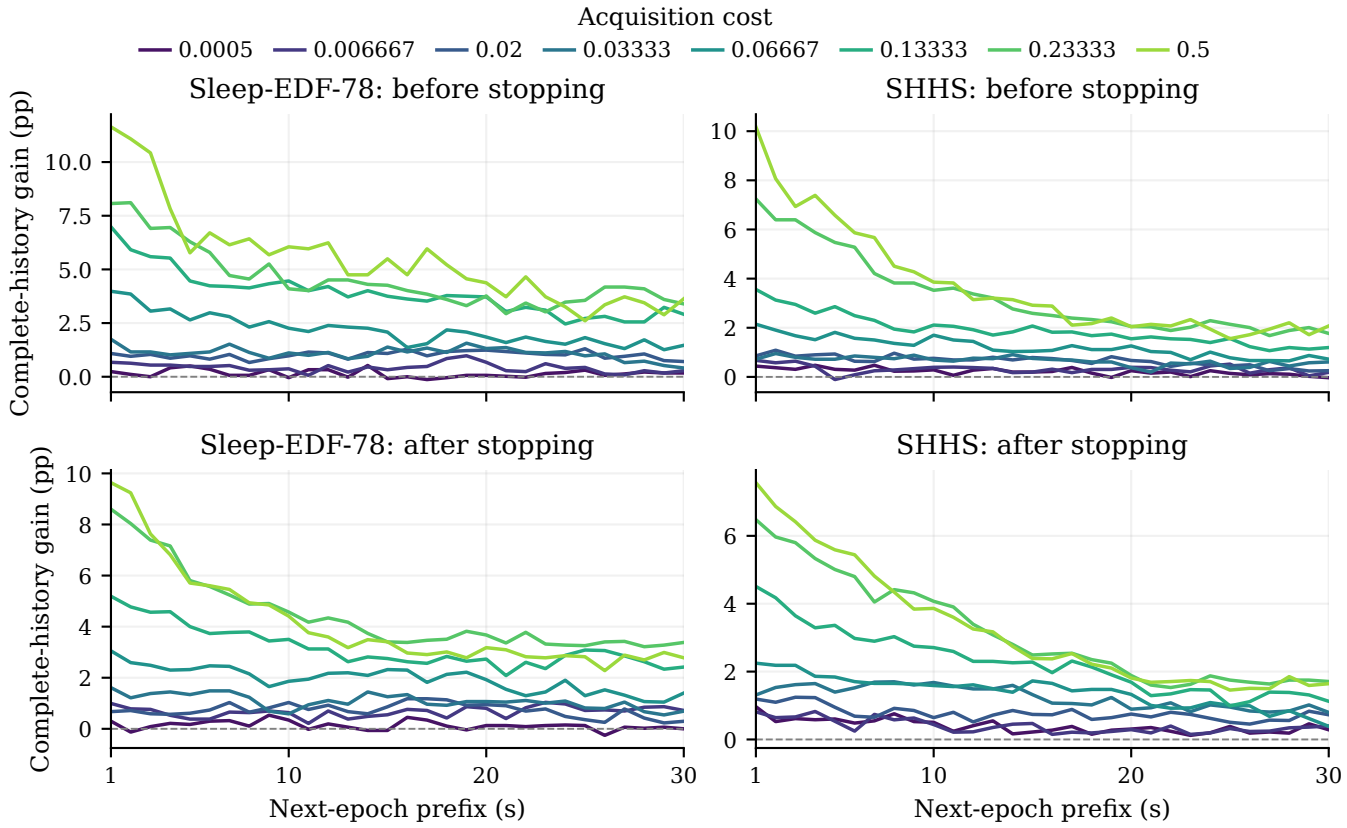



Fig. 9. Complete-history accuracy effects for selected high-change source epochs, including unmatched sources, at every next-epoch prefix and cost. The top row conditions on a high change immediately before stopping, and the bottom row on one immediately after stopping. Positive values favor the complete source endpoint. Curves are point estimates in percentage points.

the gain narrows from 2.25 points at $\eta = 0.06667$ to 0.42 at $\eta = 0.5$, consistent with less historical evidence remaining for the backbone to exploit under extreme truncation.

### C. Prefix-Wise Behavior and Accuracy

Paired-history training also strengthens the early-prefix predictions needed for adaptive acquisition (Fig. 11). At $\eta = 0.02$, Paired achieves higher fold-mean macro-F1 at all 30 prefixes on all four datasets, with one-second gains of 1.96–3.04 percentage points. At $\eta = 0.5$, one-second gains remain 2.18–3.90 points. In all eight plotted settings, the gain is larger at $k = 1$ than at $k = 30$, showing that the benefit is already present when little current-epoch EEG has been acquired.

The benefit also extends to overall classification accuracy: Paired improves prefix-averaged ACC at 31 of the 32 operating points (Table XXXV). At the tightest SHHS cost, Paired training still retains a 0.42-point macro-F1 advantage with a 0.57-point ACC decrease.

These results show that endpoint memory preserves acquired context, and paired-history training improves its use under stopper-produced histories, strengthening predictions at the short prefixes where PAUSE considers stopping.

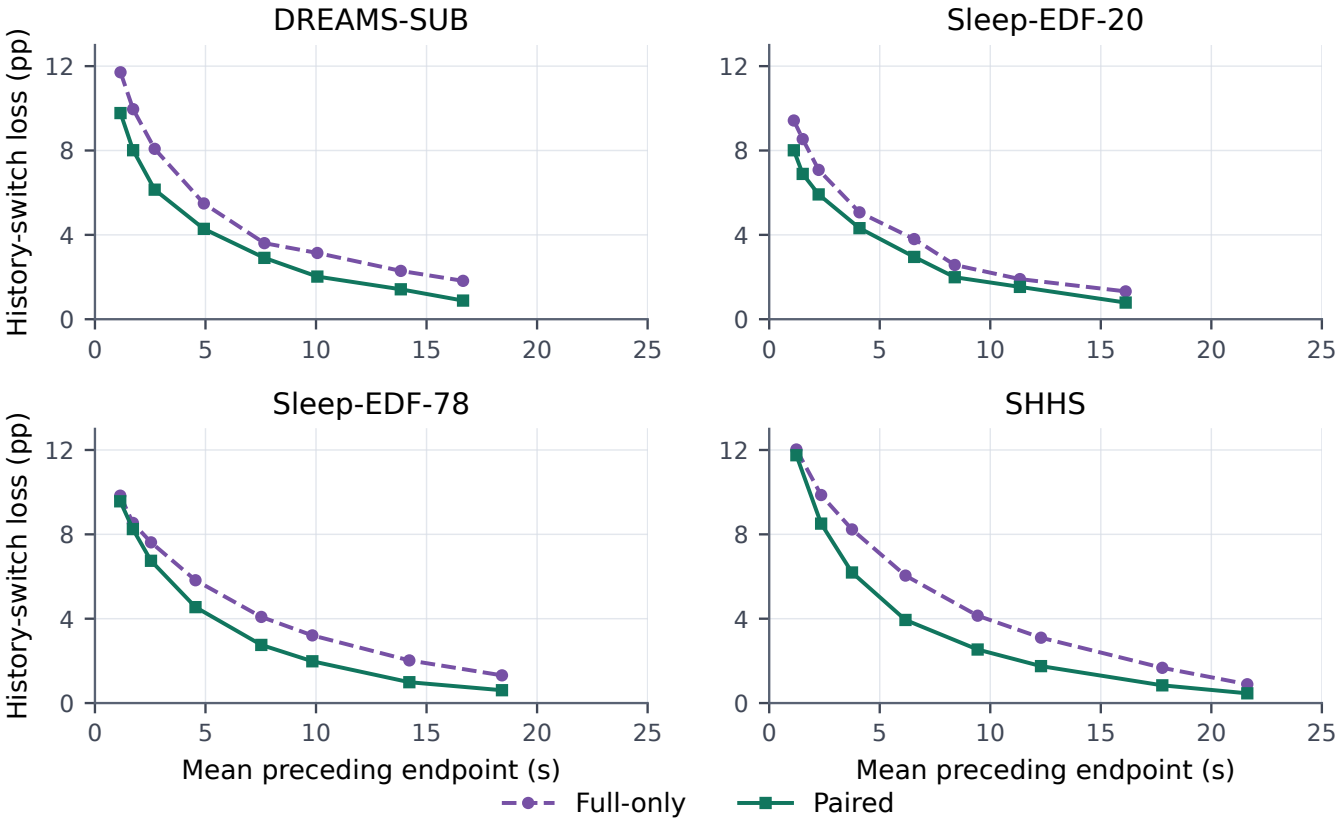



Fig. 10. Complete-minus-policy prefix-averaged macro-F1 across all eight costs. Each marker is one cost, positioned by the mean preceding endpoint actually visible to scored epochs. Vertical separation is $I$ in Eq. (7). Smaller loss indicates less sensitivity to history replacement.

TABLE XXXV
PAIRED-MINUS-FULL-ONLY PREFIX-AVERAGED ACC ON SHARED POLICY HISTORIES, IN PP (MEAN ± STANDARD DEVIATION ACROSS 20 FOLDS).

| $\eta$ | DREAMS-SUB | Sleep-EDF-20 | Sleep-EDF-78 | SHHS |
|---|---|---|---|---|
| 0.0005 | $1.51 \pm 1.56$ | $1.01 \pm 1.36$ | $1.23 \pm 1.38$ | $0.43 \pm 0.63$ |
| 0.006667 | $1.30 \pm 1.34$ | $1.03 \pm 1.79$ | $1.58 \pm 1.57$ | $0.74 \pm 0.66$ |
| 0.02 | $1.45 \pm 1.69$ | $1.15 \pm 1.99$ | $1.97 \pm 1.56$ | $1.31 \pm 0.78$ |
| 0.03333 | $1.13 \pm 1.47$ | $1.60 \pm 1.80$ | $2.15 \pm 1.73$ | $1.60 \pm 0.90$ |
| 0.06667 | $1.71 \pm 1.93$ | $1.57 \pm 1.95$ | $2.27 \pm 1.90$ | $2.11 \pm 1.19$ |
| 0.13333 | $2.14 \pm 2.55$ | $1.66 \pm 1.91$ | $1.69 \pm 2.15$ | $2.06 \pm 1.50$ |
| 0.23333 | $1.89 \pm 2.65$ | $2.13 \pm 1.97$ | $1.15 \pm 2.33$ | $1.10 \pm 1.84$ |
| 0.5 | $1.78 \pm 2.90$ | $1.99 \pm 1.94$ | $1.05 \pm 3.04$ | $-0.57 \pm 2.51$ |

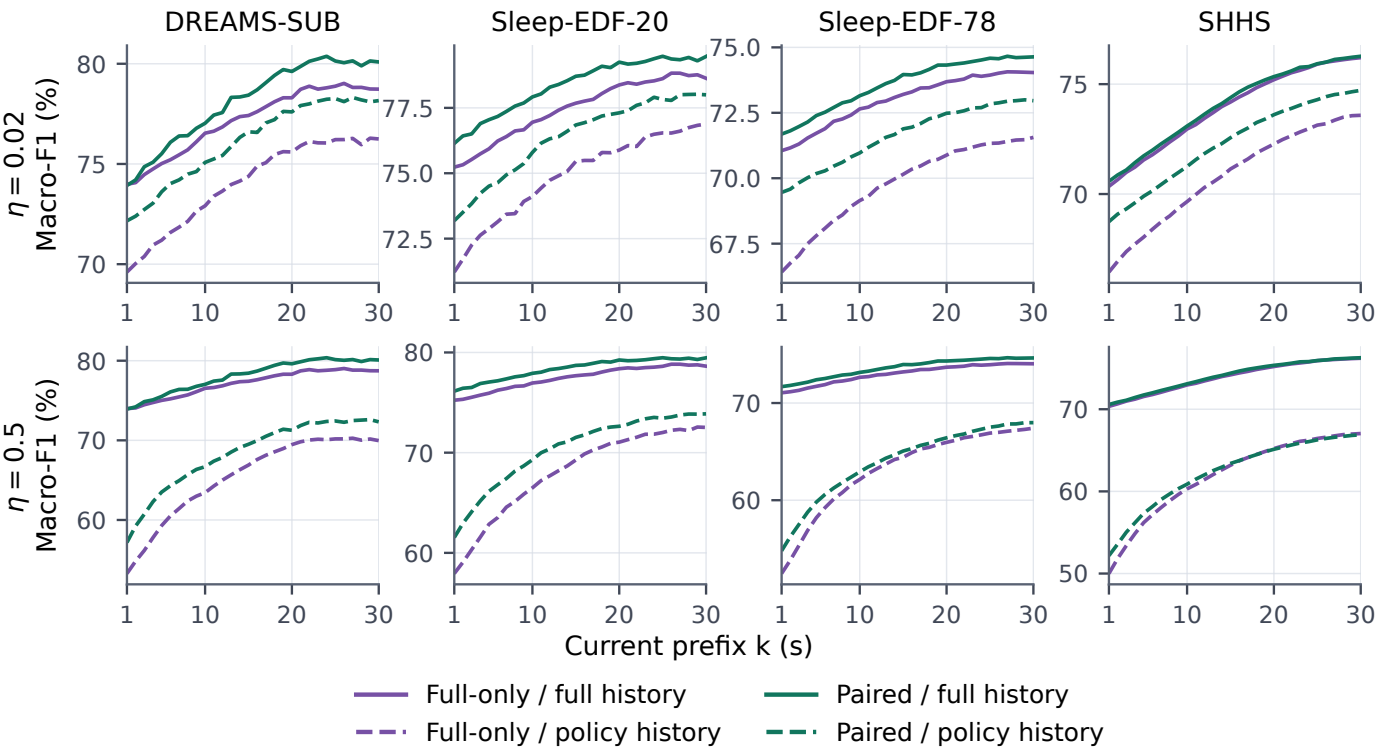



Fig. 11. Prefix-wise macro-F1 under complete and frozen policy histories at $\eta = 0.02$ (top) and $0.5$ (bottom). Colors distinguish training and line styles distinguish history. Current prefixes are evaluated offline with historical endpoints held fixed.

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